Uniform Bessel bounds for magnetization crossings in finite Ising chains

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Abstract

In a finite zero-field Ising chain, the all-up configuration and zero magnetization become equally likely at a single coupling. This crossing has recently been described as a thermal transition of the finite chain, an analogue of a first-order transition temperature, and approximated by an analytic estimate and a numerical fit. We find its asymptotics, with explicit constants, for free and for periodic ends. The coupling at the crossing grows like half the logarithm of the length of the chain, minus half the logarithm of that logarithm. So the correlation length at the crossing is a vanishing fraction of the chain. Periodic ends lower the coupling at the crossing by an amount that decays like the inverse logarithm of the length. The earlier approximations lack the doubly logarithmic term, which explains why a fit on chains of moderate length finds a slope below one half. The proofs treat the magnetization as the position of a persistent random walk and compare its central probability with a modified Bessel function, with a relative error bound that is explicit for every length. For free ends this traps the crossing in an explicit interval at every length. The proof is analytic for long chains and uses exact integer arithmetic for short ones. We also locate, for each smaller magnetization, the coupling at which it becomes as likely as full magnetization.

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1 Introduction

1.1 The crossing

Take N spins $\sigma_i = \pm 1$ in a row, with free ends and the zero-field Ising weight $\exp(J \sum_{i<N} \sigma_i \sigma_{i+1})$, where $J > 0$. Here J is dimensionless, and it is the product βJ of the inverse temperature and the exchange coupling in the notation of [16]. The all-up configuration is the only one with magnetization $\sigma_1 + \cdots + \sigma_N = N$. Zero magnetization (magnetization ± 1 for odd N) comes from very many configurations, each with a few domain walls. For small J the walls cost little and zero magnetization is the more likely of the two, and for large J the all-up configuration is. They are equally likely at a single coupling, the *central crossing*. Stepanyan et al. [16] study this crossing in the periodic chain as a thermal transition of a finite system, the analogue of the temperature of a first-order transition, and approximate it by $\beta J \approx (2 \log N - 1)/5$. We find its asymptotics with explicit constants for free and for periodic ends, and for free ends we bound it at every N.

Read from left to right, the spins are the steps of a persistent random walk [12, 11, 15]. The first step is fair, and each later step repeats the previous one with probability p, where $e^{-2J} = (1-p)/p$.

The magnetization is the position after N steps, so the walk that never turns, which reaches an end, competes with the many walks that end in the middle.

With two steps the walk ends in the middle exactly when it turns once, so the three bins have probabilities $\frac{p}{2}$, $1-p$ and $\frac{p}{2}$, and they are level at $p=\frac{2}{3}$. With three steps the bins have probabilities $\frac{p^2}{2}$, $\frac{1-p^2}{2}$, $\frac{1-p^2}{2}$ and $\frac{p^2}{2}$, and they are level at $p=1/\sqrt{2}$. In general exactly one word reaches a given endpoint, namely the word that never turns, and very many words reach the middle by turning a few times (each turn brings a factor of about $1-p$).

The middle bin and the endpoints are level when the walk turns about $\log N$ times on average. More precisely, the crossing value p_N of p satisfies $N(1-p_N)=\log N-\frac{1}{2}\log\log N-0.467\ldots+o(1)$, and for every $N\geq 2$ we trap $N(1-p_N)$ between two explicit functions of N whose distance tends to 0 (Theorem 3.4). For $p<p_N$ the middle is more likely, and for $p>p_N$ the ends are. In terms of the coupling, the crossing is at $J=\frac{1}{2}(\log N-\log\log N)+O(\log\log N/\log N)$ for free and for periodic ends, and the correlation length there is about $N/(2\log N)$ (Corollary 5.4). The approximations of [16] have no $\log\log N$ term, and this term explains the slope of their numerical fit (Table 3).

The crossing lies between two regimes in which the law is well understood. When $\tau=N(1-p)$ stays bounded, the walk is close to the telegraph process and the magnetization has a scaling limit [15, 14, 13, 4]. When p is fixed, the bin has a normal limit (Theorem 2.5). At the crossing τ grows like $\log N$, so neither limit decides it, and we need bounds that hold uniformly in τ. Our main tool is a comparison of the middle bin with a modified Bessel function, with a relative error that is explicit for every N (Theorem 3.3).

The crossing for free ends is also Problem 131 of Kagey's Open Problem Collection [1], posed for a Galton board whose ball keeps its direction with probability p. A walk with d directions, which is the d-state Potts chain, is treated in [2], and general slowly switching chains in [3].

1.2 Setting and notation

The walk. The walk makes $N\geq 1$ steps. We record a run as a word $w=w_1\cdots w_N$ in the letters $+$ (right) and $-$ (left). The letter w_1 is $+$ or $-$ with probability $\frac{1}{2}$, and given $w_1,\ldots,w_i$,

$$\mathbb{P}(w_{i+1}=w_i)=p,\qquad \mathbb{P}(w_{i+1}\neq w_i)=q=1-p.$$

An index $i<N$ with $w_i\neq w_{i+1}$ is a *switch*. Let $\mathrm{chg}(w)$ be the number of switches and $\mathrm{rep}(w)=N-1-\mathrm{chg}(w)$. Then

$$\mathbb{P}(w)=\tfrac{1}{2}\,p^{\mathrm{rep}(w)}q^{\mathrm{chg}(w)}. \tag{1}$$

For $p>0$ we put $u=q/p$, the odds of a switch.

Bins. The walk ends in bin k if w has k letters $+$, for $0\leq k\leq N$. We write $P_N(k)$ for the probability of bin k, and $P_0(0)=1$. Bins 0 and N are the *endpoints*. Only the constant word $-\cdots-$ reaches bin 0, and only $+\cdots+$ reaches bin N, so

$$P_N(0)=P_N(N)=\tfrac{1}{2}p^{N-1}\qquad (N\geq 1).$$

A middle bin is $m_N=\lfloor N/2\rfloor$. For odd N the two middle bins have equal probability by symmetry.

Spins. Put $\sigma_i=+1$ if $w_i=+$ and $\sigma_i=-1$ if $w_i=-$, and $X_N=\sigma_1+\cdots+\sigma_N$. The bin is $K_N=(N+X_N)/2$. Since $\mathrm{chg}(w)=\sum_{i<N}(1-\sigma_i\sigma_{i+1})/2$, (1) gives $\mathbb{P}(w)\propto\exp(J\sum_{i<N}\sigma_i\sigma_{i+1})$ with $e^{-2J}=u$. So the spins form the Ising chain in zero field with free boundaries, and $J>0$ exactly when $p>\frac{1}{2}$.

Scales. Let $L = \log N$ and $\tau = Nq$. Each of the $N-1$ steps after the first switches with probability q, so the expected number of switches is $(N-1)q = \tau - q$. We call τ the *switching scale*. In the Ising chain a switch is a domain wall, and $q = 1/(1+e^{2J})$, so τ is the expected number of domain walls plus q. The central crossing has τ of order L. Near the edges the crossings have larger τ, up to order $\sqrt{N}$ for the bins next to an edge (Section 4).

Ratios and crossings. For a bin k with $1 \le k \le N-1$ we write $a = k$ and $b = N-k$ for its numbers of letters $+$ and $-$. For $p > 0$ the ratio of bin k to an endpoint depends on p only through u, and it is a polynomial in u with nonnegative integer coefficients ((13) below). We write

$$S_{a,b}(u) = \frac{P_N(a)}{P_N(0)}, \qquad S_N(u) = S_{m_N,\,N-m_N}(u)$$

for this ratio and for its value at the middle. *The crossing* of bin k is the value of p at which bin k is exactly as likely as an endpoint, that is, $S_{a,b}(u) = 1$. It is unique (Section 4), and we write $p_{a,b}$ for it and $u_{a,b} = (1-p_{a,b})/p_{a,b}$ for its odds. The *central crossing* is $p_N = p_{m_N,N-m_N}$, with odds u_N, and

$$z_N = N(1-p_N) = \frac{Nu_N}{1+u_N} \tag{2}$$

is the value of τ there.

Special functions and asymptotic notation. The functions I_0 and I_1 are the modified Bessel functions of the first kind, of orders zero and one. We use them through their power series and their behavior at infinity [18, Eqs. 10.25.2, 10.40.1],

$$I_0(2x) = \sum_{r\ge0} \frac{x^{2r}}{(r!)^2}, \quad I_1(2x) = \sum_{r\ge0} \frac{x^{2r+1}}{r!\,(r+1)!}, \quad I_\nu(z) = \frac{e^z}{\sqrt{2\pi z}}\Big(1 - \frac{4\nu^2-1}{8z} + O(z^{-2})\Big)$$

as $z \to \infty$, for $\nu = 0, 1$. We write $f \sim g$ if $f/g \to 1$. Each $O(\cdot)$ comes with the parameters it is uniform in, and $O_a(\cdot)$ means that the constant may depend on a.

1.3 Main results

The coupling at the crossing. In Ising terms the free central crossing is the coupling $J_N^{\mathrm{F}} = -\frac{1}{2}\log u_N$, and J_N^{P} is the coupling at the periodic one. Put $c_0 = \frac{1}{2}\log(\pi/8) = -0.467\ldots$ and $c_{\mathrm{P}} = \frac{1}{2}\log(\pi/2)$. Theorem 5.2 and Corollary 5.4 give

$$J_N^X = \tfrac{1}{2}(\log N - \log\log N) + \frac{\frac{1}{4}\log\log N - \frac{1}{2}c_X}{\log N} + O\Big(\frac{(\log\log N)^2}{(\log N)^2}\Big)$$

for free ends ($X = \mathrm{F}$, with $c_{\mathrm{F}} = c_0$) and for periodic ends ($X = \mathrm{P}$). Since $c_{\mathrm{P}} = c_0 + \log 2$, periodic ends lower the coupling at the crossing by $(\log 2)/(2\log N)$ to leading order. At either crossing the correlation length is about $N/(2\log N)$, a vanishing fraction of the chain. The approximation $\beta J \approx (2\log N - 1)/5$ of [16, Eq. (16)] grows too slowly. At $N = 10^5$ the exact periodic coupling is 4.576, the periodic case of the display, without its error term, gives 4.578, and the approximation gives 4.405 (Table 3).

A Bessel bound at every N. Put $h = N/2$ and $F(x) = 2x(I_0(2x) + I_1(2x))$. Theorem 3.3 says that for every $N \geq 2$ and every $x > 0$,

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}.$$

Near the crossing $x = Nu/2$ is about $L/2$, so the relative error is $O(L^2/N)$ (tiny for large N). Theorem 4.3 gives the same kind of bound for every bin, with relative error at most $Nu^2/2 + u$.

The central crossing at every N. Let ζ_N be the positive root of $8\zeta e^{2\zeta} = \pi N^2$, that is, $\zeta_N = \frac{1}{2}W_0(\pi N^2/4)$ with W_0 the principal branch of Lambert's function, and let $\varepsilon_N = \zeta_N(\zeta_N + 1)/N + 1/(16\zeta_N^2)$. Theorem 3.4 says that

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N} \qquad \text{for every } N \geq 2.$$

Note that $\varepsilon_N \to 0$. For $N \geq 175$ the proof is analytic. For $2 \leq N \leq 174$ we decide the sign of $S_N - 1$ at two rationals, one on each side of the root, in exact integer arithmetic, and the Lean kernel repeats this check. The theorem has two simpler consequences. With c_0 as above, the crossing satisfies

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \qquad |R_N| \leq \frac{1}{4L^2} + \frac{L^2 + L}{N} \quad (N \geq 2)$$

(Proposition 3.8), and as $N \to \infty$

$$z_N = L - \tfrac{1}{2}\log L + c_0 + \frac{\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8}}{L} + O\Big(\frac{(\log L)^2}{L^2}\Big)$$

(Corollary 3.7). The convergence is slow, and z_N/L is still only 0.84 at $N = 10^4$ (Figure 2).

Every bin. Let bin a be at distance a from the nearer edge, with $b = N - a \geq a$. Next to the edge the crossing is at $p_N^{\mathrm{adj}} = (1 + \sqrt{N-1})/(2 + \sqrt{N-1})$. For $N \geq 4$ the two bins next to the edges are the least likely of all at the central crossing (Proposition 4.2). As $a \to \infty$, Theorem 4.7 expands $2\sqrt{ab}\,u_{a,b}$ to the same order as z_N, uniformly in $b \geq a$. For fixed a the crossing comes instead from a Laguerre polynomial. Then $u_{a,b} = c_a/\sqrt{b} - 1/b + O_a(b^{-3/2})$, where c_a^2 is the positive root of an explicit polynomial of degree a (Theorem 4.8). The two regimes match (Theorem 4.9).

Periodic boundaries. If the steps are closed into a loop, so that the last step also interacts with the first, the central crossing moves by $(\log 2)/N$ in the odds to leading order (Theorem 5.2). Under either boundary, at odds $u_N^X + t/N$, where u_N^X is the central crossing of that boundary, the ratio of bin $N/2 + yN/\sqrt{\log N}$ to an endpoint tends to e^{t-2y^2} (Theorem 5.3).

We prove these results in Sections 2 to 5, in this order. Section 6 lists what was computed and what is in Lean, and the appendix has the longer proofs of Section 3.

1.4 Related work

The correlated random walk goes back to Goldstein [12] and Gillis [11]. Renshaw and Henderson [15] study the same symmetric model, motivated by a Galton machine. They give the exact law from run counts, the generating function, the normal limit and the upturn at the endpoints, and in their

Section 4 a Bessel limit at a fixed persistence scale, followed by a limit as that scale grows. Since the two limits are taken one after the other, there is no error bound at finite N when the scale grows like $\log N$, which is where the crossing happens. Konno [14] writes the exact law with the hypergeometric function ${}_2F_1$ and proves the telegraph limit when $p = 1 - \theta/N$ with θ fixed, again a fixed scale. Herrmann and Vallois [13] give the telegraph limit with its Bessel density. Dekking and Kong [9] study the Markov binomial law, give its variance and find when it has one, two or three modes. Their criterion already gives the crossing of the bins next to the edges (Section 4.1).

For the Ising chain, Antal, Droz and Rácz [4] give the magnetization law at a fixed scale τ under several boundary conditions, and Dantchev and Rudnick [8] give exact free and periodic coefficients. The fixed-τ scaling law of Antal, Droz and Rácz already predicts the central crossing. If one evaluates it at τ of order $\log N$ (a regime in which it is not proved), its endpoint atom and its density at zero magnetization balance at a value z of τ with $z + \frac{1}{2}\log z = L + c_0 + 1/(8z) + O(z^{-2})$, which is the crossing equation of Proposition 3.8 without its remainder. What we add is a proof in this regime, with explicit bounds, and the crossing at every N. Stepanyan et al. [16, Section III.B] study two crossings of the periodic chain as thermal transitions. In the first, the all-up configuration and magnetization $N-2$ (one flipped spin) are equally likely exactly at $J = \frac{1}{4}\log N$, which they give to leading order. In the second, the all-up configuration and zero magnetization are equally likely, and they estimate the coupling. We give the second with its constants under both boundary conditions, and for free ends we bound it at every N and find the crossing of every bin. The Markov binomial estimates of Čekanavičius and Mikalauskas [6, 7] assume ranges of the transition probabilities that exclude $p \to 1$.

2 The exact law

The results of this section are classical [15, 9]. We prove Theorem 2.1 because the later sections use its count of words by switches. By (1), a word with j switches has probability $\frac{1}{2}p^{N-1-j}q^j$.

Theorem 2.1 (bin probabilities)**.** *Let $N \geq 1$. Then $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, and for $1 \leq k \leq N-1$*

$$P_N(k) = \frac{1}{2}\sum_{j=1}^{N-1} W(N,k,j)\, p^{N-1-j}q^j,$$

where, writing $a = k$ and $b = N-k$,

$$W(N,k,2i-1) = 2\binom{a-1}{i-1}\binom{b-1}{i-1}, \qquad W(N,k,2i) = \binom{a-1}{i}\binom{b-1}{i-1} + \binom{a-1}{i-1}\binom{b-1}{i}$$

is the number of words with a letters $+$, b letters $-$ and exactly j switches.

Proof. By (1) it suffices to count the words in bin k with j switches. We count them by their run lengths. A word with j switches consists of $j+1$ maximal runs, alternately of letters $+$ and of letters $-$. If $k \in \{0, N\}$ there is one word and it has no switch. Let $a, b \geq 1$. If $j+1 = 2i$ there are i runs of each letter. The run lengths form a composition of a into i parts and a composition of b into i parts, and there are $\binom{a-1}{i-1}\binom{b-1}{i-1}$ such pairs. The two choices of the first letter give the factor 2. If $j+1 = 2i+1$ the word starts and ends with the same letter. Starting with $+$ gives $i+1$ runs of $+$ and i runs of $-$, so $\binom{a-1}{i}\binom{b-1}{i-1}$ words, and starting with $-$ gives the other summand. □

For instance $P_2 = (\frac{p}{2},\, q,\, \frac{p}{2})$, $P_3(1) = \frac{1}{2}(1-p^2)$ and $P_4(2) = q(1-p+p^2)$. Each interior bin needs at least one switch, while the endpoints need none. So the endpoints win when p is close to 1 (the walk rarely turns), and Section 3 asks how close.

Proposition 2.2 (recurrence and generating function)**.** *Let $P_N^+(k)$ and $P_N^-(k)$ be the probabilities of being in bin k after N steps with last letter $+$, respectively $-$, and set them to 0 outside $0 \le k \le N$. Then $P_1^+(1) = P_1^-(0) = \frac12$, $P_1^+(0) = P_1^-(1) = 0$, and for $N \ge 2$*

$$P_N^+(k) = p\,P_{N-1}^+(k-1) + q\,P_{N-1}^-(k-1), \qquad P_N^-(k) = p\,P_{N-1}^-(k) + q\,P_{N-1}^+(k). \tag{3}$$

Consequently

$$\sum_{N\ge0}\sum_{k=0}^{N} P_N(k)\,x^k t^N = \frac{1-(p-\frac12)(1+x)\,t}{1-p(1+x)\,t+(2p-1)\,x\,t^2}, \tag{4}$$

and for all $N \ge 2$ and all k, with $P_M(k) = 0$ outside $0 \le k \le M$,

$$P_N(k) = p\big(P_{N-1}(k) + P_{N-1}(k-1)\big) - (2p-1)\,P_{N-2}(k-1). \tag{5}$$

Proof. The recursion (3) follows by conditioning on w_{N-1}, since w_N equals w_{N-1} with probability p. Let $H^+ = \sum_{N,k} P_N^+(k) x^k t^N$ and define H^- alike. By (3), $H^+ = xt(\frac12 + pH^+ + qH^-)$ and $H^- = t(\frac12 + pH^- + qH^+)$. Solving these two linear equations and adding 1 for $N = 0$ gives (4), which is the generating function of Renshaw and Henderson [15] in our notation. Multiplying (4) by its denominator and comparing the coefficients of $x^k t^N$ with $N \ge 2$ gives (5). □

Remark 2.3 (numerator tables). Problem 131 of [1] tabulates the bins for $p = \frac23$ and $p = \frac13$. Row n of those tables has n bins, so it is our $N = n-1$. For $p = \alpha/(\alpha+\beta)$ with integers $\alpha, \beta \ge 0$, not both 0, the numerators of the tables are $n_N(k) = 2(\alpha+\beta)^{N-1} P_N(k)$ for $N \ge 1$, not reduced. By (1), $n_N(k)$ is the sum of $\alpha^{\mathrm{rep}(w)}\beta^{\mathrm{chg}(w)}$ over the words with k letters $+$ and $N-k$ letters $-$, which shows that it is an integer.

2.1 Variance and the normal limit

Note that K_N counts the visits of the chain $w_1, \dots, w_N$ to the letter $+$. This chain leaves each letter with probability q and starts from its stationary law $(\frac12, \frac12)$. So K_N has the Markov binomial law $\mathrm{Bin}(N, q, q, (\frac12, \frac12))$ of Dekking and Kong [9], whose second eigenvalue is $1 - 2q = 2p - 1$.

Proposition 2.4 (mean and variance)**.** *Let $0 < p < 1$ and $\rho = 2p - 1$. Then $\mathbb{E}K_N = N/2$ and*

$$\operatorname{Var} K_N = \frac14\Big(N\,\frac{1+\rho}{1-\rho} - \frac{2\rho\,(1-\rho^N)}{(1-\rho)^2}\Big) = \frac{Np}{4(1-p)} - \frac{(2p-1)\big(1-(2p-1)^N\big)}{8(1-p)^2}.$$

Proof. This is [9, Eq. (2.3) and Proposition 2.1] with $a = b = q$ and the stationary start, for which both excentricities vanish and both stationary weights are $\frac12$. The second form follows from $(1+\rho)/(1-\rho) = p/q$ and $1 - \rho = 2q$. □

Theorem 2.5 (normal limit)**.** *Let $0 < p < 1$ be fixed. Then $(K_N - N/2)/\sqrt{N}$ converges in distribution, as $N \to \infty$, to the normal law with mean 0 and variance $p/(4(1-p))$.*

Proof. Let $v_N(\theta)$ be the vector with entries $\mathbb{E}[e^{i\theta X_N}; \sigma_N = \pm1]$. By (3), $v_{N+1} = M v_N$ with

$$M = \begin{pmatrix} pe^{i\theta} & qe^{i\theta} \\ qe^{-i\theta} & pe^{-i\theta} \end{pmatrix}, \qquad \operatorname{tr} M = 2p\cos\theta, \quad \det M = 2p - 1.$$

At $\theta = 0$ the eigenvalues are 1 and $2p-1$, which are distinct. So for small $|\theta|$ the matrix M has two simple eigenvalues. One is $\lambda_+(\theta) = p\cos\theta + \sqrt{q^2 - p^2\sin^2\theta} = 1 - \frac{p}{2q}\theta^2 + O(\theta^4)$, and the other,

$\lambda_-(\theta)$, stays near $2p-1$. Since both are simple, the spectral projections $\Pi_\pm(\theta)$ depend continuously on θ. Write $\mathbf{1} = (1,1)$. Then $\mathbb{E}e^{i\theta X_N} = \mathbf{1}\cdot M^{N-1}v_1(\theta) = \lambda_+^{N-1}\,\mathbf{1}\cdot\Pi_+v_1 + \lambda_-^{N-1}\,\mathbf{1}\cdot\Pi_-v_1$. The vector $v_1(0) = (\frac12,\frac12)$ is an eigenvector of M for the eigenvalue 1 at $\theta = 0$, so $\mathbf{1}\cdot\Pi_+v_1 \to 1$ as $\theta\to 0$, while $|\lambda_-| \to |2p-1| < 1$. Put $\theta = s/\sqrt{N}$ with s fixed. Then the second term tends to 0, and the first tends to $\exp(-\frac{p}{2q}s^2)$ since $(N-1)\log\lambda_+(s/\sqrt{N}) \to -\frac{p}{2q}s^2$. By Lévy's continuity theorem [5, Theorem 26.3], $X_N/\sqrt{N}$ converges to the normal law with variance p/q. Finally $K_N - N/2 = X_N/2$. □

For $p > \frac12$ persistence widens the bell by the factor $\sqrt{p/q}$ compared with the simple random walk, and for $p < \frac12$ it narrows it by the same factor.

3 The central crossing at every N

Let $N \geq 2$, let $m_N = \lfloor N/2\rfloor$ and let $p > 0$. By Theorem 2.1 the ratio of the middle bin to an endpoint is

$$\frac{P_N(m_N)}{P_N(0)} = S_N(u) := \sum_{j=1}^{N-1} W(N, m_N, j)\, u^j, \tag{6}$$

a polynomial in u with nonnegative integer coefficients and $W(N, m_N, 1) = 2$. The middle and the endpoints are level exactly when $S_N(u) = 1$.

We first give a heuristic for why this happens when τ is near $\log N$. Let $a = m_N$ and $b = N - m_N$, both about $N/2$. A word in the middle bin with $2r+1$ switches has $r+1$ runs of each letter, and there are $2\binom{a-1}{r}\binom{b-1}{r} \approx 2(N/2)^{2r}/(r!)^2$ of them when r is small compared with $\sqrt{N}$. So these words contribute about $2u\sum_r (Nu/2)^{2r}/(r!)^2 = 2uI_0(Nu)$ to $S_N(u)$, and in the same way the words with an even number of switches contribute about $2uI_1(Nu)$. For large $w = Nu$ we have $I_0(w) + I_1(w) \approx 2e^w/\sqrt{2\pi w}$, so $S_N(u) \approx 4we^w/(N\sqrt{2\pi w})$. Setting this equal to 1 gives $w + \frac12\log w \approx L + \frac12\log(\pi/8)$. So at the crossing w, and with it τ, is close to $L - \frac12\log L + c_0$ with $c_0 = \frac12\log(\pi/8)$.

Theorem 3.1. *(a) For every $N \geq 2$ there is exactly one $p = p_N \in [0,1]$ with $P_N(m_N) = P_N(0)$, and $p_N \geq \frac23$. The middle is more likely than the endpoints for $p < p_N$ and less likely for $p > p_N$.*

(b) The first values are $p_2 = \frac23$, $p_3 = 1/\sqrt{2}$ and the real root p_4 of $3p^3 - 4p^2 + 4p - 2$, and $p_N < p_{N+1}$ for all $N \geq 2$.

(c) As $N\to\infty$ we have $p_N \to 1$, and more precisely $1 - p_N \sim (\log N)/N$.

Proof. (a) At $p = 0$ the endpoints have probability 0 and the middle does not. For $p > 0$ the equation is $S_N(u) = 1$ by (6). The coefficients of S_N are nonnegative and the linear one is 2, so S_N is strictly increasing on $[0,\infty)$ with $S_N(0) = 0$ and $S_N(u) \geq 2u$. By the intermediate value theorem there is exactly one root $u_N \in (0, \frac12]$. The map $u \mapsto p = 1/(1+u)$ is a decreasing bijection from $[0,\infty)$ to $(0,1]$. Hence $p_N = 1/(1+u_N) \geq \frac23$, and the middle wins exactly when $u > u_N$.

(b) By Theorem 2.1, $S_2 = 2u$, $S_3 = 2u + u^2$ and $S_4 = 2u + 2u^2 + 2u^3$, which give the three values. For p_4, substitute $u = (1-p)/p$ in $S_4(u) = 1$. For the monotonicity we compare coefficients. Going from N to $N+1$, one entry of the middle pair $(a,b) = (\lfloor N/2\rfloor, \lceil N/2\rceil)$ grows by one and the other is unchanged. The formulas of Theorem 2.1 are symmetric in (a,b), and every $\binom{a-1}{\cdot}$ is nondecreasing in a. So $W(N+1, m_{N+1}, j) \geq W(N, m_N, j)$ for every j. The coefficient of u^2 is $a + b - 2 = N - 2$, which increases strictly. Hence $S_{N+1}(u) > S_N(u)$ for $u > 0$, so $u_{N+1} < u_N$.

(c) We prove (c) after Theorem 3.3, from part (i) of that theorem. □

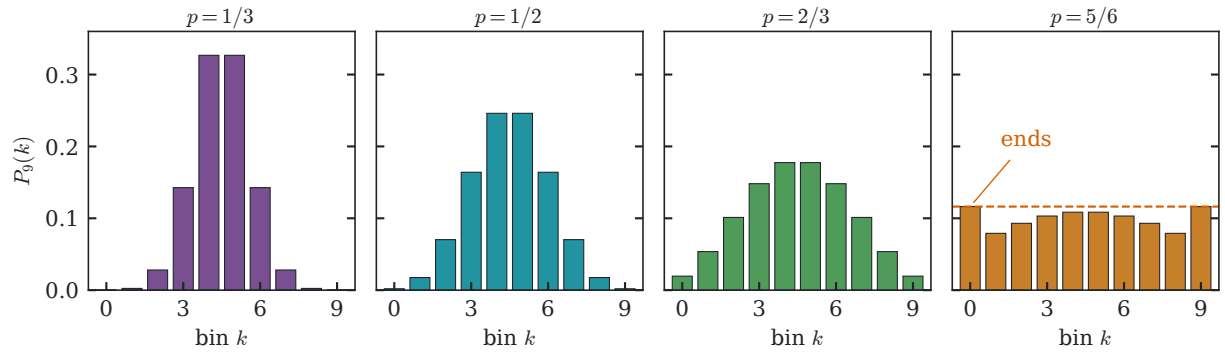


Figure 1: The bin probabilities $P_9(k)$ of the tenth row for $p = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$. At $p = \frac{5}{6}$, just above p_9, the two middle bins are slightly below the endpoints (dashed line).

In the tables of [1] (Remark 2.3), the middle and the endpoints of row $N+1$ are level at $p = p_N$. Among the odd rows from the third on the least value is $p_2 = \frac{2}{3}$, and among the even rows from the fourth on it is $p_3 = 1/\sqrt{2}$. (The first row has one bin, and the second is $1, 1$ for every p.) There is no greatest value, since $p_N \uparrow 1$. The tenth row has $p_9 = 0.82829\ldots$, just below $\frac{5}{6}$, so its middle and endpoints are nearly level in the third histogram of [1] (Figure 1 and Table 1).

N	p_N	N	p_N	N	p_N	N	p_N
2	0.66666667	6	0.78871211	10	0.83840411	50	0.94200405
3	0.70710678	7	0.80361731	11	0.84652725	100	0.96496633
4	0.74487450	8	0.81759938	12	0.85426060	200	0.97939018
5	0.76759188	9	0.82829163	20	0.89293837	400	0.98812464

Table 1: The value p_N at which the middle and endpoint bins of row $N+1$ are equally likely.

3.1 A Bessel bound at every N

The heuristic above replaced the number $\binom{a-1}{r}$ of ways to split a letters into $r+1$ runs by $a^r/r!$. After this change the run count is exactly the power series of I_0 and I_1. The next lemma bounds the error this makes.

Lemma 3.2 (clipped products)**.** *(a) If $t_1, \ldots, t_r \geq 0$ and $(v)_+ = \max(v, 0)$, then $0 \leq 1 - \prod_{k=1}^{r}(1 - t_k)_+ \leq \sum_{k=1}^{r} t_k$. (b) For integers $m \geq 1$ and $r \geq 0$, $U_r(m) := \frac{r!}{m^r}\binom{m-1}{r} = \prod_{k=1}^{r}\left(1 - \frac{k}{m}\right)_+$, so $0 \leq 1 - U_r(m) \leq \frac{r(r+1)}{2m}$. (c) For $x \geq 0$,*

$$\sum_{r\geq 0} \frac{r(r+1)x^{2r}}{(r!)^2} = x^2 I_0(2x) + x I_1(2x),$$

$$\sum_{r\geq 0} \frac{r(r+1)x^{2r+1}}{r!(r+1)!} = x^2 I_1(2x), \qquad \sum_{r\geq 0} \frac{(r+1)x^{2r+1}}{r!(r+1)!} = x I_0(2x).$$

Proof. (a) Every factor lies in $[0, 1]$, so the product is at most 1. If some $t_k \geq 1$ the product is 0 and the sum is at least 1. Otherwise $\prod_k (1 - t_k) \geq 1 - \sum_k t_k$ by induction on r. (b) Note that $\frac{r!}{m^r}\binom{m-1}{r} = \prod_{k=1}^{r} \frac{m-k}{m}$. For $r < m$ every factor is positive, and for $r \geq m$ both sides vanish, the

right side through the factor $k = m$. The bound is (a) with $t_k = k/m$. (c) Use the series of $I_0(2x)$ and $I_1(2x)$ from Section 1.2. Write $r(r+1) = r^2 + r$ in the first sum, cancel the factors r and $r+1$ against the factorials, and shift the index. □

Theorem 3.3 (uniform Bessel approximation)**.** *Put $h = N/2$ and $F(x) = 2x\big(I_0(2x) + I_1(2x)\big)$, and recall that $u_N = (1-p_N)/p_N$.*

(i) For every $N \geq 2$ and $x > 0$,

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}. \tag{7}$$

(ii) Let $x_N = hu_N$, let x_B be the unique positive root of $F(x_B) = h$, and put $\delta_N = x_N(x_N+1)/h$. Then $x_B \leq x_N$, and $x_N - x_B \leq -\frac{1}{2}\log(1-\delta_N)$ whenever $\delta_N < 1$.

(iii) With $u_B = x_B/h$ we have $0 \leq u_N - u_B = O(L^2/N^2)$ as $N \to \infty$.

Section 3.2 uses part (i) to locate p_N at every N.

Proof. (i) *Step 1. The clipped factors.* Let $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$. As in Lemma 3.2(b), $A_r = \frac{r!}{h^r}\binom{a-1}{r} = \prod_{k=1}^{r}(1 - \frac{h-a+k}{h})_+$, and B_r is the same with b in place of a. Since $a+b = 2h$ and $h-a \in \{0, \frac{1}{2}\}$, the $2r$ numbers $(h-a+k)/h$ and $(h-b+k)/h$ are nonnegative with sum $r(r+1)/h$, so Lemma 3.2(a) gives $0 \leq 1 - A_rB_r \leq r(r+1)/h$. In the same way the deficits of $A_{r+1}B_r$ and A_rB_{r+1} are at most $((r+1)^2 + h - a)/h$ and $((r+1)^2 + h - b)/h$. Averaging cancels the parity offsets (the shifts $h-a$ and $h-b$ add to 0), so $0 \leq 1 - \frac{1}{2}(A_{r+1}B_r + A_rB_{r+1}) \leq (r+1)^2/h$.

Step 2. Summing. With $u = x/h$ the run-count formula of Theorem 2.1 becomes

$$S_N(u) = 2u\Big[\sum_{r\geq 0} \frac{x^{2r}}{(r!)^2} A_rB_r + \sum_{r\geq 0} \frac{x^{2r+1}}{r!(r+1)!} \frac{A_{r+1}B_r + A_rB_{r+1}}{2}\Big].$$

Without the factors A_r, B_r the bracket is $I_0(2x)+I_1(2x)$ and the right side is $F(x)/h$. By Step 1 and Lemma 3.2(c), $F(x)/h - S_N(u)$ lies between 0 and $2u/h$ times $\sum_r r(r+1)\frac{x^{2r}}{(r!)^2} + \sum_r (r+1)^2 \frac{x^{2r+1}}{r!(r+1)!} = x(x+1)\big(I_0(2x) + I_1(2x)\big)$. Dividing by $F(x)/h = 2u(I_0(2x) + I_1(2x))$ proves (i). This covers every r, including those beyond the degree of the polynomial, where the clipped products vanish and Lemma 3.2(a) still applies.

(ii) Differentiating the series term by term gives $F' = 2F + 2I_0(2x)$. So F increases from 0 to ∞, x_B is unique, and $(\log F)' \geq 2$. At $x = x_N$ we have $hS_N(u_N) = h$, so (i) gives $h \leq F(x_N)$, hence $x_B \leq x_N$, and $F(x_N) \leq h/(1-\delta_N)$ when $\delta_N < 1$. Integrating $(\log F)' \geq 2$ from x_B to x_N gives $2(x_N - x_B) \leq \log(F(x_N)/h) \leq -\log(1-\delta_N)$.

Proof of Theorem 3.1(c). We use only part (i). Note that $x^{2r}/(r!)^2 \leq (2x)^{2r}/(2r)!$ and $x^{2r+1}/(r!(r+1)!) \leq (2x)^{2r+1}/(2r+1)!$ for $x \geq 0$, since $\binom{2r}{r} \leq 4^r$ and $\binom{2r+1}{r} \leq 2^{2r+1}$. So $I_0(2x) + I_1(2x) \leq e^{2x}$ and $F(x) \leq 2xe^{2x}$. Fix $\theta \in (0,1)$ and put $x_\pm = (1 \pm \theta)L/2$. Then $F(x_-) \leq (1-\theta)LN^{1-\theta} < h$ for large N, so $hS_N(x_-/h) < h$ by (i), and $x_- < x_N$ since S_N increases. On the other side $F(x_+) \geq 2x_+I_0(2x_+)$, which is at least a constant times $\sqrt{L}\,N^{1+\theta}$ for large N by the large-argument form of I_0, while $x_+(x_+ + 1)/h \to 0$. So $hS_N(x_+/h) \geq F(x_+)(1 - x_+(x_+ + 1)/h) > h$ for large N, and $x_N < x_+$. Hence $Nu_N = 2x_N \sim L$, so $u_N \to 0$ and $z_N = Nu_N/(1+u_N) \sim L$, which is (c).

(iii) By Theorem 3.1(c), $x_N \sim L/2$. So $\delta_N = O(L^2/N)$, and (ii) gives $x_N - x_B = O(L^2/N)$. Divide by h. □

3.2 The crossing at every N

We measure the crossing by $z_N = N(1-p_N) = Nu_N/(1+u_N)$, the value of τ at $p = p_N$. Let $\psi(z) = z + \frac{1}{2}\log z$ and $c_0 = \frac{1}{2}\log(\pi/8) = -0.46735\ldots$, and let ζ_N be the positive root of

$$\psi(\zeta_N) = L + c_0, \quad \text{that is,} \quad 8\zeta_N e^{2\zeta_N} = \pi N^2 \quad \text{or} \quad \zeta_N = \tfrac{1}{2}W_0(\pi N^2/4), \tag{8}$$

with W_0 the principal branch of the Lambert function. This is the heuristic equation from the start of the section with w replaced by ζ. Since $\zeta e^{2\zeta}$ increases, ζ_N increases with N. Also $\zeta_N \le L$, since $\psi(L) \ge L + c_0$ when $L \ge \pi/8$. Finally let $\varepsilon_N = \zeta_N(\zeta_N+1)/N + 1/(16\zeta_N^2)$.

Theorem 3.4 (the crossing at every N)**.** *For every $N \ge 2$,*

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N}.$$

The width ε_N tends to 0, since $\zeta_N \le L$ and $\zeta_N \to \infty$. It is 0.440, 0.163, 0.0387 and 0.00778 at $N = 10, 10^2, 10^3, 10^4$ (Table 2).

The two terms ζ_N and $1/(8\zeta_N)$ come from the following heuristic. Let $\Phi(w) = F(w/2) = w\big(I_0(w) + I_1(w)\big)$ and $\mathcal{E}(z) = \sqrt{\pi z/2}\, e^{-z}\big(I_0(z) + I_1(z)\big)$, which tends to 1 as $z \to \infty$. Put $w_N = Nu_N$ and $\Lambda_N = 2\Phi(w_N)/N$. At the crossing, (7) and (9) below give $\psi(w_N) + \log\mathcal{E}(w_N) = L + c_0 + \log\Lambda_N$ with $0 \le \log\Lambda_N \le -\log(1 - w_N(w_N+2)/(2N))$ whenever $w_N(w_N+2) < 2N$. With $\mathcal{E} = \Lambda_N = 1$ this would say $w_N = \zeta_N$. Since $\log\mathcal{E}(w) = -1/(8w) + O(w^{-2})$ and $z_N = w_N + O(w_N^2/N)$, it suggests $z_N = \zeta_N + 1/(8\zeta_N) + O(\zeta_N^{-2} + \zeta_N^2/N)$. The theorem makes this error explicit and one-sided. Its proof uses two lemmas. Lemma 3.5 turns a comparison of z_N with a test value into a sign condition, and Lemma 3.6 bounds $\mathcal{E}$ with explicit constants.

Lemma 3.5 (root bracket)**.** *Let $N \ge 2$ and $w > 0$. Then*

$$\log\Phi(w) - \log\tfrac{N}{2} = \psi(w) - \psi(\zeta_N) + \log\mathcal{E}(w). \tag{9}$$

(a) If $\Phi(w) \le N/2$, then $Nw/(N+w) \le z_N$, strictly if $\Phi(w) < N/2$. (b) If $\Phi(w)\big(1 - \frac{w(w+2)}{2N}\big) > N/2$, then $z_N < Nw/(N+w)$.

Proof. Note that $\log\Phi(w) = \psi(w) + \frac{1}{2}\log\frac{2}{\pi} + \log\mathcal{E}(w)$ and $\log\frac{N}{2} - \psi(\zeta_N) = -\log 2 - c_0 = \frac{1}{2}\log\frac{2}{\pi}$, which gives (9). Now apply (7) with $x = w/2$. Then $F(x) = \Phi(w)$ and $x(x+1)/h = w(w+2)/(2N)$, so

$$\Phi(w)\Big(1 - \frac{w(w+2)}{2N}\Big) \le \frac{N}{2}\, S_N\Big(\frac{w}{N}\Big) \le \Phi(w).$$

(a) The hypothesis gives $S_N(w/N) \le 1 = S_N(u_N)$, strictly if $\Phi(w) < N/2$. Since S_N is strictly increasing, $w/N \le u_N$, strictly in that case. The map $v \mapsto Nv/(1+v)$ increases and sends w/N to $Nw/(N+w)$ and u_N to z_N. (b) Now $S_N(w/N) > 1$, so $w/N > u_N$. □

Lemma 3.6 (Bessel bounds)**.** *For every $z > 0$,*

$$1 - \frac{1}{8z} - \frac{3}{128z^2} - \frac{45}{512z^3} \le \mathcal{E}(z) \le 1 - \frac{1}{8z} - \frac{3}{128z^2} + \frac{15}{512z^3},$$

and $\mathcal{E}(z) \le 1$. In particular $\mathcal{E}(z) \le 1 - \frac{1}{8z}$ for $z \ge \frac{5}{4}$.

The lemma is the large-argument expansion of $I_0 + I_1$ with explicit errors. We prove it in Appendix A. By [18, Eq. 10.32.3] and a substitution, $\mathcal{E}(z) = \pi^{-1/2}\int_0^{2z} e^{-v}v^{-1/2}\sqrt{1 - v/(2z)}\,dv$. We bound $\sqrt{1-y}$ above and below by cubics that share its Taylor terms up to y^2, extend the integral to ∞, and use $\Gamma(k+\frac{1}{2})/\Gamma(\frac{1}{2}) = 1, \frac{1}{2}, \frac{3}{4}, \frac{15}{8}$ for $k = 0, 1, 2, 3$.

Proof of Theorem 3.4. Step 1. The split at $N = 175$. By (8), $N = \sqrt{8\zeta_N/\pi}\, e^{\zeta_N}$, and the right side increases in ζ_N. Since $\sqrt{32/\pi}\, e^4 = 174.25\ldots$, we have $\zeta_N \geq 4$ exactly when $N \geq 175$. Write $\zeta = \zeta_N$ and $U = \zeta + \frac{1}{8\zeta}$.

Step 2. The upper bound for $N \geq 175$. We apply Lemma 3.5(b) at $W = NU/(N-U)$, for which $NW/(N+W) = U$. By (9) it suffices that $G_U = \psi(W) - \psi(\zeta) + \log \mathcal{E}(W) + \log(1 - W(W+2)/(2N)) > 0$. Expanding each term with Lemma 3.6 gives $G_U \geq \frac{1}{32\zeta^2} + \frac{\zeta(\zeta-1)}{2N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, and Appendix A makes every term explicit for $\zeta \geq 4$.

Step 3. The lower bound for $N \geq 175$. We apply Lemma 3.5(a) at $W' = NU'/(N-U')$ with $U' = U - \varepsilon_N$. It suffices that $G_{U'} = \psi(W') - \psi(\zeta) + \log \mathcal{E}(W') < 0$, and we get $G_{U'} \leq -\frac{3}{128\zeta^2} - \frac{\zeta}{N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, again with explicit constants for every $\zeta \geq 4$ in Appendix A. The term $-\zeta/N$ is why ε_N contains $\zeta(\zeta+1)/N$.

Step 4. Short chains, by certificates. Here $2 \leq N \leq 174$. Let $d = 10^9$, let $c = \lfloor du_N \rfloor$ and let $c' = c+1$. At $N = 2$, where $du_2 = 5 \cdot 10^8$ is an integer, we take $c = du_2 - 1$ and $c' = c+2$. The sign of $S_N(c/d) - 1$ is the sign of $n_N(m_N) - d^{N-1}$, where $n_N(m_N) = d^{N-1} S_N(c/d)$ is the integer of Remark 2.3 at $(\alpha, \beta) = (d, c)$ (Appendix A.3). We computed these integers at c and c' by two exact methods, the run-count sum of Theorem 2.1 and the recursion (3) with integer weights d, c in place of p, q. They agreed for every $N \leq 174$ and gave $c/d < u_N < c'/d$, so $z(c/d) < z_N < z(c'/d)$ with $z(u) = Nu/(1+u)$. We then checked $\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z(c/d)$ and $z(c'/d) < \zeta_N + \frac{1}{8\zeta_N}$ in 60-digit decimal arithmetic, with ζ_N in a bracket of width 10^{-45}. We certified this bracket by the sign of $\psi(\zeta) - L - c_0$, which is the sign of $8\zeta e^{2\zeta} - \pi N^2$, at both of its ends, using a correctly rounded logarithm. The smallest margins are 0.033193 (upper) and 0.085591 (lower), both at $N = 174$. The following estimate shows that rounding cannot change these comparisons. Put $g_1(\zeta) = \zeta + \frac{1}{8\zeta}$ and $g_2(\zeta) = g_1(\zeta) - \frac{\zeta(\zeta+1)}{N} - \frac{1}{16\zeta^2}$. For $\frac{1}{2} \leq \zeta \leq L+1$ and $N \geq 2$ we have $|g_1'|, |g_2'| \leq 5$, and $\frac{1}{2} < \zeta_2 \leq \zeta_N \leq L$. So g_1 and g_2 vary by at most $5 \cdot 10^{-45}$ on the bracket, and the decimal evaluation adds at most 10^{-50}. Both errors are far below the smallest margin, which exceeds 0.03, so the comparisons hold. Lean checks them again with rational bounds on ζ_N and π. We also ran the certificates up to $N = 1000$ as a sanity check (both methods up to 400, then only the run-count sum), and they all pass. □

N	z_N	ζ_N	$z_N - \zeta_N - \frac{1}{8\zeta_N}$	$-\varepsilon_N$	R_N
2	0.666666666667	0.5368290974	−0.1030112	−0.62938	+0.050643
10	1.615958871376	1.6001732978	−0.0623310	−0.44048	−0.056660
100	3.503367368092	3.5100055489	−0.0422507	−0.16337	−0.043265
174	3.996779704730	3.9987131703	−0.0331935	−0.11878	−0.033450
175	4.001967989170	4.0038072816	−0.0330596	−0.11838	−0.033304
1000	5.590351166975	5.5807388611	−0.0127862	−0.03873	−0.011887
10^4	7.734265514999	7.7210118331	−0.0029359	−0.00778	−0.002051

Table 2: The crossing against Theorem 3.4, and the remainder R_N of (12). We computed z_N by run-count bisection in 60-digit decimal arithmetic and by a floating-point dynamic program, which agree to the digits shown.

The bracket is in terms of ζ_N (a value of the Lambert function). The next corollary states it for p_N and expands it in $L = \log N$.

Corollary 3.7. *Put $\nu_N = L - \frac{1}{2}\log L + c_0$.*

(a) For every $N \geq 2$, $1 - \big(\zeta_N + \frac{1}{8\zeta_N}\big)/N < p_N < 1 - \big(\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N\big)/N$.

(b) For every $N \ge 2$, $\nu_N \le \zeta_N \le \nu_N + \frac{\log L - 2c_0}{4\nu_N}$ and

$$-\varepsilon_N < z_N - \nu_N < \frac{\log L - 2c_0 + \frac{1}{2}}{4\nu_N}. \tag{10}$$

In particular $z_N = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$ as $N \to \infty$.

(c) As $N \to \infty$,

$$z_N = \nu_N + \frac{\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8}}{L} + O\Big(\frac{(\log L)^2}{L^2}\Big). \tag{11}$$

(d) (Computer-assisted.) $z_N > \zeta_N$ for every $N \ge 2774$.

Proof. (a) is Theorem 3.4 with $p_N = 1 - z_N/N$.

(b) By (8), $\zeta_N = L + c_0 - \frac{1}{2}\log\zeta_N$. Since $\zeta_N \le L$, this gives $\zeta_N \ge \nu_N$, and $\nu_N > 0$ since it increases in N and $\nu_2 = 0.41$. Also $\log y \le y - 1$ gives $\zeta_N - \nu_N = \frac{1}{2}\log(L/\zeta_N) \le (L - \zeta_N)/(2\zeta_N)$, and $L - \zeta_N = \frac{1}{2}\log\zeta_N - c_0 \le \frac{1}{2}\log L - c_0$. Hence $\zeta_N - \nu_N \le (\log L - 2c_0)/(4\nu_N)$. For z_N, add Theorem 3.4 and $0 < \frac{1}{8\zeta_N} \le \frac{1}{8\nu_N}$. The last claim follows since $\varepsilon_N = O(L^2/N + L^{-2})$.

(c) Put $\mu = \zeta_N - \nu_N = \frac{1}{2}\log(L/\zeta_N)$. By (b), $L - \zeta_N = \frac{1}{2}\log L - c_0 + O(\log L/L)$, so $\mu = \frac{L-\zeta_N}{2L} + O\big(\frac{(\log L)^2}{L^2}\big) = \frac{\frac{1}{4}\log L - \frac{1}{2}c_0}{L} + O\big(\frac{(\log L)^2}{L^2}\big)$. Also $\frac{1}{8\zeta_N} = \frac{1}{8L} + O(\frac{\log L}{L^2})$ and $\varepsilon_N = O(L^2/N + L^{-2})$, and Theorem 3.4 gives (c).

(d) By Theorem 3.4, $z_N - \zeta_N > \frac{1}{8\zeta_N} - \varepsilon_N$, which is positive when $f(\zeta_N) > 1$, where $f(\zeta) = n(\zeta)\big(\frac{1}{8\zeta} - \frac{1}{16\zeta^2}\big)/(\zeta(\zeta+1))$ and $n(\zeta) = \sqrt{8\zeta/\pi}\,e^{\zeta}$, so that $n(\zeta_N) = N$. The logarithmic derivative of f is $1 + \frac{1}{2\zeta} - \frac{2}{\zeta} + \frac{1}{2\zeta^2-\zeta} - \frac{1}{\zeta+1} > 1 - \frac{3}{2\zeta} - \frac{1}{\zeta+1} > 0$ for $\zeta \ge 3$. Since ζ_N increases with N, once $f(\zeta_N) > 1$ it stays above 1. This step is computer-assisted. In 60-digit decimal arithmetic, with ζ_N bracketed within 10^{-45} as in Step 4 above, we found $f(\zeta_{2774}) - 1 = 2.1 \times 10^{-4}$, and the sign holds on the whole bracket. Since $\zeta_{2774} > 3$, this proves (d). The condition fails at $N = 2773$. □

The heuristic suggests that $\psi(z_N) \approx L + c_0 + 1/(8z_N)$. The next proposition proves this at every N with an explicit remainder.

Proposition 3.8 (the crossing equation)**.** *For every $N \ge 2$,*

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \qquad -\frac{z_N^2 + z_N}{N} \le R_N \le \frac{1}{16z_N^2}. \tag{12}$$

We also have $L/2 \le z_N \le L$, so $|R_N| \le \frac{1}{4L^2} + \frac{L^2+L}{N}$ for every $N \ge 2$.

We prove Proposition 3.8 in Appendix A. The proof writes R_N exactly, using (7) and (9) at $w = w_N$ (formula (41)). For $N \ge 175$, Lemma 3.6 puts the Bessel part of this formula between 0 and $0.0559/z_N^2$, and the other terms lie between $-(z_N^2 + z_N)/N$ and 0. For $N \le 174$ the proof uses the same certificates as Theorem 3.4.

Remark 3.9 (the sign of $z_N - \zeta_N$, computer-assisted). We have $z_N > \zeta_N$ for $N \le 15$, $z_N < \zeta_N$ for $16 \le N \le 218$, and $z_N > \zeta_N$ for every $N \ge 219$. For $N \le 1000$ the certificates decide the sign, since ζ_N never lies in $[z(c/d), z(c'/d)]$. The smallest distance is 1.66×10^{-5}, at $N = 218$. Both exact methods cover $N \le 400$, and the run-count sum alone covers the rest. For $1001 \le N \le 2773$ we checked $S_N(v) < 1$ in exact integers from the run-count sum, at a rational v above $\zeta_N/(N - \zeta_N)$ by at least 1.2×10^{-11}, where $z(\zeta_N/(N - \zeta_N)) = \zeta_N$. These integers were cross-checked by floating-point run-count and dynamic-program sign tests for every $N \le 3000$, and by the integer recursion at $N = 401, 1000, 1008, 2000, 2773$. For $N \ge 2774$ the sign is Corollary 3.7(d).

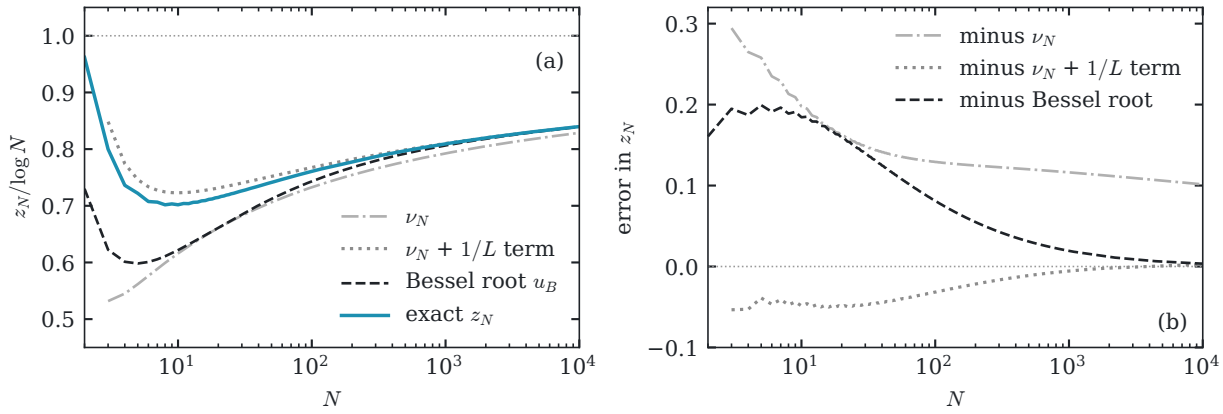


Figure 2: (a) The scaled crossing $z_N/\log N$ (solid) and the same quantity from the Bessel root u_B of Theorem 3.3 (dashed), from ν_N of (10) (through the constant) and from $\nu_N + (\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8})/L$ of (11) (with the $1/L$ term). The ratio tends to 1 very slowly. (b) The exact z_N minus each approximation.

The ratios u_B/u_N of the Bessel root of Theorem 3.3 to the true odds are 0.9761, 0.9864, 0.9924 at $N = 100, 200, 400$ and 0.99579, 0.99769, 0.99874 at $N = 800, 1600, 3200$, in line with the relative error $O(L^2/N)$ of Theorem 3.3. Figure 2 compares the exact crossing with the expansions of Corollary 3.7 up to $N = 10^4$.

4 The crossing of every bin

For each bin k we find the value of p at which bin k is exactly as likely as an endpoint. Figure 3 shows it for three values of N. The crossing $1 - p_{a,N-a}$ decreases as the bin moves in from the edge, from order $N^{-1/2}$ next to the edge to order $(\log N)/N$ in the middle.

4.1 Ordering the crossings

Let $a, b \geq 1$ be integers, $N = a + b$ and $p > 0$, and recall that $S_{a,b}(u) = P_N(a)/P_N(0)$ with $u = (1-p)/p$. Theorem 2.1 and $\binom{m-1}{r+1} = \frac{m-1-r}{r+1}\binom{m-1}{r}$ for $m = a, b$ give

$$S_{a,b}(u) = \sum_{r\geq 0} \binom{a-1}{r}\binom{b-1}{r}\Big(2u^{2r+1} + \frac{N-2-2r}{r+1}u^{2r+2}\Big). \tag{13}$$

Its coefficients are nonnegative, as $N - 2 - 2r \geq |b - a|$ for $r < \min(a, b)$, and the linear one is 2. So there is a unique $u_{a,b} > 0$ with $S_{a,b}(u_{a,b}) = 1$, and $p_{a,b} = 1/(1 + u_{a,b})$. As $S_{a,b}(u) \geq 2u$, we have $u_{a,b} \leq \frac{1}{2}$ and $p_{a,b} \in [\frac{2}{3}, 1)$, with $p_{a,b} = \frac{2}{3}$ only for $a = b = 1$.

Proposition 4.1 (ordering the crossings)**.** *If $a < b - 1$, then*

$$u_{a+1,b-1} < u_{a,b}, \qquad p_{a+1,b-1} > p_{a,b}.$$

Also $p_{a,b+1} > p_{a,b}$ for all positive a, b. In addition, $S_{a+1,b-1} \geq S_{a,b}$ and $S_{a,b+1} \geq S_{a,b}$ coefficientwise, with at least one strict coefficient.

So a bin closer to the middle, or in a longer row, needs a larger p to fall to the endpoint height.

Proof. For fixed N and $r \le a-1$, $\binom{a-1}{r}\binom{b-1}{r} = (r!)^{-2}\prod_{j=0}^{r-1}(a-1-j)(b-1-j)$, a product of positive factors. Each factor grows when (a,b) becomes $(a+1,b-1)$, by $(a-j)(b-2-j)-(a-1-j)(b-1-j) = b-a-1 > 0$. The even multiplier $(N-2-2r)/(r+1)$ is unchanged, and it is nonnegative where either product is nonzero, since then $r \le a$ and $N-2-2r \ge b-a-2 \ge 0$. New summands are nonnegative, and the cubic coefficient grows strictly, from $2(a-1)(b-1)$ to $2a(b-2)$. Increasing b with a fixed increases the binomial coefficients of Theorem 2.1, and the quadratic coefficient $N-2$ strictly. A polynomial with nonnegative coefficients that dominates another coefficientwise, strictly somewhere, is strictly larger at every $u > 0$, so its root is smaller. □

The ordering of the bin heights in Proposition 4.2 also follows from the interior log-concavity of Dekking and Kong [9, Proposition 4.3]. We use the coefficients because they also give the strict gap in that proposition. The formula for p_N^{adj} in that proposition is already implicit in their work. By [9, Theorem 4.1], for $N \ge 4$ and $0 < p < 1$ the row is trimodal exactly when $P_N(0) > P_N(1)$, since $P_N(1) < P_N(2)$ here, and their Example 4.2 with $a = b = q$ and the stationary start gives $P_N(1)/P_N(0) = 2u + (N-2)u^2$, which is $S_{1,N-1}(u)$. The next proposition adds the comparison $p_N > p_N^{\mathrm{adj}}$ for $N \ge 4$ and the global extrema at p_N.

Proposition 4.2 (adjacent bins and global extrema)**.** *For $N \ge 2$ and $0 < p < 1$, an adjacent bin has the same probability as an endpoint precisely at*

$$p_N^{\mathrm{adj}} = \frac{1+\sqrt{N-1}}{2+\sqrt{N-1}}.$$

For $0 < p < 1$, the endpoints are the least likely bins if $p < p_N^{\mathrm{adj}}$, bins 1 and $N-1$ are if $p > p_N^{\mathrm{adj}}$, and they tie at equality. For $N \ge 4$, the central crossing satisfies $p_N > p_N^{\mathrm{adj}}$. At $p = p_N$, the middle bin and the endpoints attain the global maximum and bins $1, N-1$ attain the global minimum. For $N = 2, 3$, $p_N = p_N^{\mathrm{adj}}$ and every bin ties at the crossing.

So at the central crossing the row has the shape of a W. It is high at the two ends and in the middle, and lowest next to the ends.

Proof. By (13), $S_{1,N-1}(u) = 2u + (N-2)u^2$, with positive root $u^{\mathrm{adj}} = 1/(1+\sqrt{N-1})$. This gives p_N^{adj} and the sign change. For $0 < p < 1$ we have $P_N(k) = P_N(0)S_{k,N-k}(u)$ with $P_N(0) > 0$, so if $1 \le k < N-k-1$ Proposition 4.1 gives $P_N(k) < P_N(k+1)$. By symmetry the middle is the largest interior bin and bins $1, N-1$ are the smallest, which gives the claims about the least likely bins. For $N \ge 4$ the cubic coefficient $2(\lfloor N/2 \rfloor - 1)(\lceil N/2 \rceil - 1)$ of S_N is positive, so $2u_N + (N-2)u_N^2 < S_N(u_N) = 1$, hence $u_N < u^{\mathrm{adj}}$ and $p_N > p_N^{\mathrm{adj}}$. At p_N the middle ties with the endpoints, which gives the global maximum, and bins $1, N-1$ are below the endpoints, which gives the global minimum. For $N = 2, 3$ we have $S_N = S_{1,N-1}$, and symmetry gives the ties. □

So $1 - p_N^{\mathrm{adj}} \sim N^{-1/2}$, much larger than $1 - p_N \sim (\log N)/N$. Away from the ends, the single letter + of bin 1 needs two switches (one into it and one out of it), so $S_{1,N-1} \approx Nu^2$ reaches 1 only at $u \approx N^{-1/2}$. (At $p = 0$ and $p = 1$ there are also ties between bins of probability 0.)

4.2 A Bessel estimate uniform in the bin

Bin a has about $(a^r/r!)(b^r/r!)$ words with r runs of each letter, each of weight about u^{2r}. The sum depends only on abu^2, so a bin near the edge acts like the middle of a shorter row with $\sqrt{ab}$

in place of $N/2$. Put

$$s=\sqrt{ab},\quad A=\frac{ab}{a+b},\quad c=\frac{a+b}{2\sqrt{ab}},\quad \eta=\frac{1}{c},\quad x=su,\quad \mathcal{J}_c(x)=I_0(2x)+cI_1(2x),$$

and $\widehat{S}_{a,b}(u)=2u\mathcal{J}_c(su)$. Here $c\ge 1$ and $0<\eta\le 1$, with $\eta=1$ exactly when $a=b$.

Theorem 4.3 (uniform estimate for every bin)**.** *For all positive integers a,b and all $u>0$,*

$$0\le 1-\frac{S_{a,b}(u)}{\widehat{S}_{a,b}(u)}\le \frac{(a+b)u^2}{2}+u=\frac{x^2}{2A}+\frac{x}{s}. \tag{14}$$

In particular, for fixed N and u the bound does not depend on the bin. The inequality still holds when the right side exceeds 1, but then it gives no lower bound on $S_{a,b}$.

When u is about L/N, the bound is $O(L^2/N)$ for every bin at once. The proof follows that of Theorem 3.3(i), with the two letters weighted by a and b.

Proof. Let $U_r=U_r(a)$ and $V_r=U_r(b)$ as in Lemma 3.2(b), and note that $1/a+1/b=1/A$. Lemma 3.2(a), applied to the $2r$ clipped factors of U_rV_r, gives

$$0\le 1-U_rV_r\le \frac{r(r+1)}{2A}. \tag{15}$$

Lemma 3.2(a) gives $1-U_{r+1}V_r\le \frac{(r+1)(r+2)}{2a}+\frac{r(r+1)}{2b}$, and the same with a and b exchanged for U_rV_{r+1}. Averaging with weights a and b, the weighted average $E_r=(aU_{r+1}V_r+bU_rV_{r+1})/(a+b)$ satisfies

$$\begin{aligned}0\le 1-E_r&\le \frac{(r+1)(r+2)}{a+b}+\frac{r(r+1)}{2(a+b)}\Big(\frac{a}{b}+\frac{b}{a}\Big)\\&=\frac{r(r+1)}{2A}+\frac{2(r+1)}{a+b}.\end{aligned} \tag{16}$$

Since $\binom{a-1}{r}=a^rU_r/r!$, (13) becomes

$$S_{a,b}(u)=2u\Big[\sum_{r\ge 0}\frac{x^{2r}}{(r!)^2}U_rV_r+c\sum_{r\ge 0}\frac{x^{2r+1}}{r!(r+1)!}E_r\Big], \tag{17}$$

and dropping U_rV_r and E_r gives $\widehat{S}_{a,b}(u)$. So by Lemma 3.2(c), with $I_j=I_j(2x)$, $(\widehat{S}_{a,b}-S_{a,b})/(2u)$ lies between 0 and

$$\frac{1}{2A}(x^2I_0+xI_1+cx^2I_1)+\frac{2c\,xI_0}{a+b}=\Big(\frac{x^2}{2A}+\frac{x}{s}\Big)\mathcal{J}_c(x),$$

since $c/s=1/(2A)$ and $2c/(a+b)=1/s$. Divide by $\widehat{S}_{a,b}/(2u)=\mathcal{J}_c(x)$. □

To turn this into a bound on the crossing, let $x_{a,b}=su_{a,b}$ and let x_B be the positive root of

$$G_\eta(x_B)=A,\qquad G_\eta(x)=x\{I_1(2x)+\eta I_0(2x)\}. \tag{18}$$

Differentiating term by term gives

$$G_\eta'(x)-2G_\eta(x)=2x(1-\eta)\{I_0(2x)-I_1(2x)\}+\eta I_0(2x)\ge 0, \tag{19}$$

since $I_0(2x)-I_1(2x)=\frac{1}{\pi}\int_0^\pi e^{2x\cos\theta}(1-\cos\theta)\,d\theta>0$ by [18, Eq. 10.32.3]. So $(\log G_\eta)'\ge 2$ for $x>0$ and $0\le\eta\le 1$, and x_B is unique.

Corollary 4.4 (quantitative crossing bound)**.** *We have $x_{a,b} \ge x_B$. If*

$$\delta_{a,b} = \frac{x_{a,b}^2}{2A} + \frac{x_{a,b}}{s} < 1,$$

then

$$0 \le x_{a,b} - x_B \le -\tfrac{1}{2}\log(1-\delta_{a,b}). \tag{20}$$

Proof. Since $c\eta = 1$ and $2c/s = 1/A$, we have $\widehat{S}_{a,b}(u) = G_\eta(su)/A$. At the crossing $S_{a,b} = 1$, so Theorem 4.3 gives $A \le G_\eta(x_{a,b}) \le A/(1-\delta_{a,b})$. As $G_\eta(x_B) = A$, this gives $x_{a,b} \ge x_B$, and integrating $(\log G_\eta)' \ge 2$ from x_B to $x_{a,b}$ gives (20). This is the argument of Theorem 3.3(ii). For even N and $a = b$ the two statements coincide, since then $\eta = 1$, $s = h$ and $2G_1 = F$. □

Remark 4.5 (Jacobi polynomials). The run sums of this paper can be written in terms of Jacobi polynomials, and their Bessel and Laguerre forms are limits of Jacobi polynomials near the point 1. For $a \le b$ and $0 \le t < 1$,

$$\sum_{r\ge 0}\binom{a-1}{r}\binom{b-1}{r}t^r = {}_2F_1(1-a,1-b;1;t) = (1-t)^{a-1}P_{a-1}^{(0,b-a)}\Big(\frac{1+t}{1-t}\Big),$$

and for $b > a$ and $0 \le u < 1$,

$$\sum_{r\ge 0}\binom{a-1}{r}\binom{b-1}{r+1}u^{2r+2} = \frac{(b-1)u^2}{a}(1-u^2)^{a-1}P_{a-1}^{(1,b-a-1)}\Big(\frac{1+u^2}{1-u^2}\Big).$$

Near 1, Jacobi polynomials of large degree behave like Bessel functions (Mehler–Heine [18, Eq. 18.11.5]), with expansions in modified Bessel functions uniform near 1 by Elliott [10, Section 3]. For fixed a and $y = bu^2$, as $b \to \infty$, [18, Eq. 18.7.21] turns the last sum into $(y/a)L_{a-1}^{(1)}(-y) = H_a(y)$ of (25), and the other terms tend to 0. Temme [17, Section 4.2] also compares Jacobi with Laguerre polynomials. Elliott's expansions, however, are for fixed parameters and are formal, with error bounds left to later work [10, Section 1]. Here $b-a$ can be as large as $N-2$, and Theorem 4.3 gives an explicit relative error at every N.

4.3 Every growing distance from an edge

The Bessel root satisfies an equation of the form $z + \frac{1}{2}\log z = \ell + C + B/z + \dots$. The next lemma inverts such equations.

Lemma 4.6 (inverting $z+\frac{1}{2}\log z$)**.** *Fix $K > 0$. There is a constant C_K with the following property. Let $z \ge 1$ and $\ell \ge 2$, and suppose that $z + \frac{1}{2}\log z = \ell + C + B/z + \omega$ with $|C|, |B| \le K$ and $|\omega| \le 1$. Then*

$$\Big|z - \ell + \tfrac{1}{2}\log\ell - C - \frac{\frac{1}{4}\log\ell - \frac{1}{2}C + B}{\ell}\Big| \le C_K\Big(\frac{(\log\ell)^2}{\ell^2} + |\omega|\Big).$$

Proof. Put $w = z - \ell$. As $z \ge 1$, $\log z \ge 0$ and $|B/z| \le K$, so $z \le \ell + 2K + 1$ and $|w| \le 2K + 1 + \frac{1}{2}\log(\ell + 2K + 1) = O_K(\log\ell)$. This proves the claim for ℓ bounded in terms of K, since then both sides are $O_K(1)$ and $(\log\ell)^2/\ell^2$ is bounded below. For larger ℓ we have $|w| \le \ell/2$, so $\log z = \log\ell + w/\ell + O((w/\ell)^2)$ and $B/z = B/\ell + O_K(|w|/\ell^2)$, and $w = C - \frac{1}{2}\log\ell - w/(2\ell) + B/\ell + \omega + O_K((\log\ell)^2/\ell^2)$. This gives first $w = d + O_K(\log\ell/\ell + |\omega|)$ with $d = C - \frac{1}{2}\log\ell$, and then, putting this back into the term $w/(2\ell)$, $w = d + (B - d/2)/\ell + O_K((\log\ell)^2/\ell^2 + |\omega|)$. □

Theorem 4.7 (uniform crossing expansion)**.** *Let $1 \le a \le b$, and put*

$$\ell = \log A, \qquad C(\eta) = \tfrac{1}{2}\log(8\pi) - \log(1+\eta), \qquad B(\eta) = \frac{3-\eta}{8(1+\eta)}.$$

As $a \to \infty$, uniformly over $b \ge a$,

$$2\sqrt{ab}\,u_{a,b} = \ell - \tfrac{1}{2}\log\ell + C(\eta) + \frac{\frac{1}{4}\log\ell - \frac{1}{2}C(\eta) + B(\eta)}{\ell} + O\left(\frac{(\log\ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \tag{21}$$

That is, there are a_0 and K' such that the error term is at most K' times the bracket for all $a \ge a_0$ and all $b \ge a$. The same holds with $1 - p_{a,b}$ in place of $u_{a,b}$.

Here $A = ab/(a+b)$ is half the harmonic mean of a and b. There is no condition on how a compares with N, so the theorem covers a bin at distance $\sqrt{N}$ from the edge as well as the middle. The proof uses Corollary 4.4 to pass to the Bessel root x_B. The large-argument form of I_0 and I_1 then turns (18) into an equation that Lemma 4.6 solves.

Proof. Here $a/2 \le A \le a$, $s \ge a$ and $0 < \eta \le 1$.

Step 1. The crossing is at $x_{a,b} \le \log a$. At $x = \log a$ the bound of Theorem 4.3 is at most $2(\log a)^2/a$, while $\widehat{S}_{a,b} = G_\eta(x)/A \ge xI_1(2x)/a \to \infty$, both uniformly in b (the identity $\widehat{S}_{a,b}(u) = G_\eta(su)/A$ is in the proof of Corollary 4.4). So $S_{a,b} > 1$ there, and since $S_{a,b}$ increases, $x_{a,b} \le \log a$ for all $a \ge a_0$ and all $b \ge a$. Then $\delta_{a,b} \le 2(\log a)^2/a$, and Corollary 4.4 gives

$$x_{a,b} - x_B = O((\log a)^2/a). \tag{22}$$

Step 2. The Bessel root. Write $z = 2x_B$, so $z \to \infty$ as $A \to \infty$, uniformly in η. By the large-argument form of I_0 and I_1 [18, Eq. 10.40.1], $I_1(z)+\eta I_0(z) = (1+\eta)e^z(2\pi z)^{-1/2}(1-B(\eta)/z+O(z^{-2}))$ uniformly in $0 \le \eta \le 1$. Taking logarithms in (18) gives $z + \frac{1}{2}\log z = \ell + C(\eta) + B(\eta)/z + O(z^{-2})$. As $|C(\eta)| \le 2$ and $0 < B(\eta) \le \frac{3}{8}$, Lemma 4.6 with $K = 2$ expands z.

Step 3. Back to the crossing. By (22) the expansion moves to $2x_{a,b} = 2\sqrt{ab}\,u_{a,b}$. Finally $2s(u - u/(1+u)) \le 2x^2/s = O((\log a)^2/a)$ at $u = u_{a,b}$, which gives the same expansion for $1 - p_{a,b} = u_{a,b}/(1+u_{a,b})$. □

Hence, uniformly in b,

$$1 - p_{a,b} \sim \frac{\log a}{2\sqrt{ab}} \qquad (a \to \infty,\ b \ge a). \tag{23}$$

If $a = o(b)$, then $\eta \to 0$ and $\log A = \log a + o(1)$, so

$$2\sqrt{ab}(1 - p_{a,b}) = \log a - \tfrac{1}{2}\log\log a + \tfrac{1}{2}\log(8\pi) + o(1). \tag{24}$$

If $a/N \to \alpha \in (0, \frac{1}{2}]$ at any rate, then

$$2\sqrt{a(N-a)}(1 - p_{a,N-a}) = \log N - \tfrac{1}{2}\log\log N + \log\{\alpha(1-\alpha)\} + C(2\sqrt{\alpha(1-\alpha)}) + o(1).$$

At the center, $a = b = N/2$ gives $A = N/4$ and $\eta = 1$, and changing from $\log A$ to $\log N$ recovers (11).

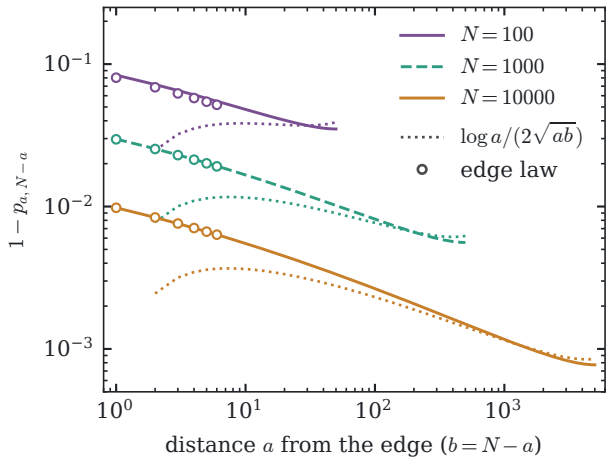


Figure 3: The crossing $1 - p_{a,N-a}$ of each bin against its distance a from the nearer edge, for $N = 100$, 1000 and 10^4. The dotted lines are the leading law (23), and the circles are the edge law (27) for $a \le 6$. The right end of each line is the central crossing $1 - p_N$.

4.4 Fixed bins and the Laguerre edge law

Now let a be fixed and $b \to \infty$. A run of letters + away from the ends needs two switches, which gives a factor u^2, and it can go in about b places. So the right variable is $y = bu^2$. Define

$$H_a(y) = \sum_{r=0}^{a-1} \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = \frac{y}{a} L_{a-1}^{(1)}(-y), \tag{25}$$

with the standard Laguerre polynomial $L_{a-1}^{(1)}$. Let y_a be the positive root of $H_a(y_a) = 1$ and $c_a = \sqrt{y_a}$. Then $y_1 = 1$, $y_2 = \sqrt{3} - 1$, and y_a decreases strictly in a, by coefficient comparison.

Theorem 4.8 (fixed-edge expansion)**.** *For each fixed positive integer a, as $b \to \infty$,*

$$u_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1}{b} + \frac{\gamma_a}{b^{3/2}} + O_a(b^{-2}), \tag{26}$$

where

$$\gamma_a = \frac{1 + y_a + (y_a + y_a^2/2) H_a''(y_a)/H_a'(y_a)}{2c_a}.$$

Thus the second coefficient in the odds expansion is -1 for every fixed bin. For the switching probability,

$$1 - p_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1 + c_a^2}{b} + O_a(b^{-3/2}). \tag{27}$$

For bin 1, where $y_1 = 1$ and $H_1'' = 0$, this gives $u_{1,b} = b^{-1/2} - b^{-1} + b^{-3/2} + \dots$, the expansion of the exact root $1/(1 + \sqrt{b})$ of Proposition 4.2.

Proof. Put $\theta = b^{-1/2}$, $y = bu^2$ and $V_r = U_r(b)$. For $b \ge a$, (13) reads

$$\begin{aligned} S_{a,b}(u) = 2\sqrt{y/b} \sum_r \binom{a-1}{r} \frac{y^r}{r!} V_r \\ + \frac{y}{b} \sum_r \binom{a-1}{r+1} \frac{y^r}{r!} V_r + y \sum_r \binom{a-1}{r} \frac{y^r}{(r+1)!} V_{r+1}. \end{aligned} \tag{28}$$

These sums have at most a terms, and with $V_r = 1$ they are $H_a'(y)$, $H_a''(y)$ and $H_a(y)$. Expanding $V_r = 1 - r(r+1)/(2b) + O_a(b^{-2})$ and using $\sum_r (r+1)(r+2)\binom{a-1}{r}\frac{y^{r+1}}{(r+1)!} = y^2 H_a''(y) + 2yH_a'(y)$ gives, uniformly near $y = y_a$,

$$\begin{aligned} S_{a,b}(u) = H_a(y) + 2\sqrt{y}\, H_a'(y)\theta \\ + \{(y - y^2/2)H_a''(y) - yH_a'(y)\}\theta^2 + O_a(\theta^3). \end{aligned} \tag{29}$$

With $b = \theta^{-2}$ the right side of (28) is analytic in (θ, y) near $(0, y_a)$, and $H_a'(y_a) > 0$. So the implicit-function theorem gives an analytic solution $y(\theta) = y_a + d_1\theta + d_2\theta^2 + O_a(\theta^3)$ of $S_{a,b} = 1$, and by the uniqueness of the crossing it is the value of y at $u = u_{a,b}$. Matching coefficients in (29) gives $d_1 = -2c_a$ and $d_2 = 2 + y_a + (y_a + y_a^2/2)H_a''(y_a)/H_a'(y_a)$. Expanding $u = \theta\sqrt{y}$ and then $u/(1+u)$ proves both expansions. □

The next theorem shows that for large a the Laguerre root y_a obeys the growing-distance law (24).

Theorem 4.9 (matching the edge and Bessel regimes)**.** *For every positive integer a and every $x > 0$,*

$$0 \le 1 - \frac{aH_a(x^2/a)}{xI_1(2x)} \le \frac{x^2}{2a}. \tag{30}$$

Consequently, with $\ell = \log a$ and $C(0) = \frac{1}{2}\log(8\pi)$ as in Theorem 4.7, as $a \to \infty$,

$$\begin{aligned} 2\sqrt{ay_a} = \ell - \tfrac{1}{2}\log\ell + C(0) \\ + \frac{\frac{1}{4}\log\ell - \frac{1}{2}C(0) + 3/8}{\ell} + O\left(\frac{(\log\ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \end{aligned} \tag{31}$$

Proof. With $U_r = U_r(a)$, $aH_a(x^2/a) = x\sum_{r\ge 0} U_r x^{2r+1}/(r!(r+1)!)$, so Lemma 3.2(b) and the second identity of Lemma 3.2(c) give (30). Put $x_a = \sqrt{ay_a}$, so $aH_a(x_a^2/a) = a$, and $x_a < \log a$ for large a by (30). So $a \le G_0(x_a) \le a/(1 - x_a^2/(2a))$, and as in Corollary 4.4, $x_a - x_0 = O((\log a)^2/a)$, where $G_0(x_0) = a$. So $z = 2x_0$ satisfies $z + \frac{1}{2}\log z = \ell + C(0) + 3/(8z) + O(z^{-2})$ by [18, Eq. 10.40.1], as in the proof of Theorem 4.7 with $\eta = 0$ and $A = a$, and Lemma 4.6 gives the expansion. □

Figure 3 compares these laws with the exact crossings.

4.5 The moving boundary layer

The crossings also give the shape of a whole row at odds $u = v_N > 0$. Between the central crossing (v_N about $(\log N)/N$) and the adjacent one (v_N about $N^{-1/2}$), the middle is above the endpoints and a layer of bins next to each edge is below them. The width of this layer is

$$\mathcal{L}_N = \#\{1 \le k \le \lfloor N/2 \rfloor : P_N(k) < P_N(0)\}.$$

Corollary 4.10 (boundary-layer width)**.** *Suppose that, as $N \to \infty$,*

$$\Omega_N = \frac{1}{\sqrt{N}\, v_N} \longrightarrow \infty, \qquad \frac{\log\Omega_N}{Nv_N} \longrightarrow 0.$$

Then

$$\mathcal{L}_N \sim \Omega_N^2 \log^2\Omega_N = \frac{\log^2(1/(\sqrt{N}\, v_N))}{Nv_N^2}. \tag{32}$$

In particular, if $v_N = N^{-\kappa}$ with $1/2 < \kappa < 1$, then

$$\mathcal{L}_N \sim (\kappa - \tfrac{1}{2})^2 N^{2\kappa-1} \log^2 N.$$

By symmetry, the total number of interior bins below the endpoint height is $2\mathcal{L}_N$ for all sufficiently large N.

Proof. Set $d_N = \Omega_N^2 \log^2 \Omega_N$. Then $d_N \to \infty$ and $d_N/N = (\log \Omega_N/(Nv_N))^2 \to 0$. Fix $\theta \in (0,1)$ and integers $a_\pm = (1 \pm \theta)d_N + O(1)$. Then $\log a_\pm \sim 2 \log \Omega_N$ and $\sqrt{a_\pm(N - a_\pm)}\, v_N \sim \sqrt{1 \pm \theta}\, \log \Omega_N$, so (23) gives $u_{a_\pm, N-a_\pm}/v_N \to 1/\sqrt{1 \pm \theta}$. Bin k is below the endpoint exactly when $v_N < u_{k,N-k}$, and $u_{k,N-k}$ decreases strictly for $k \le N/2$ by Proposition 4.1. So $a_- \le \mathcal{L}_N < a_+$ for large N, and we let $\theta \downarrow 0$. Finally $\mathcal{L}_N = o(N)$, so the reflected intervals are disjoint. □

5 Periodic boundaries and the crossing window

We now close the sequence of steps into a loop, so that the last step also interacts with the first. With this closing bond the spins form the periodic Ising chain, with weight proportional to $\exp(J \sum_{i=1}^N \sigma_i \sigma_{i+1})$, $\sigma_{N+1} = \sigma_1$ and again $u = e^{-2J}$. Antal, Droz and Rácz [4] give its magnetization law at a fixed scale, and Dantchev and Rudnick [8] give exact coefficients. We find how much the closing bond moves the central crossing, and then describe the row near the crossing under both boundaries.

Write $S^{\mathrm{F}}_{N,k} = S_{k,N-k}$ and $S^{\mathrm{P}}_{N,k}$ for the bin-to-endpoint ratios under free and periodic boundaries. Put $a = k$, $b = N - k$ and $s = \sqrt{ab}$. Counting cyclic words by their runs gives

$$S^{\mathrm{P}}_{N,k}(u) = \sum_{r=1}^{\min(a,b)} \frac{N}{r} \binom{a-1}{r-1} \binom{b-1}{r-1} u^{2r}. \tag{33}$$

To see this, note that a cyclic word with a letters $+$ and b letters $-$ switches an even number $2r$ of times and has r runs of each letter. Choose compositions of a and of b into r positive runs and a position for the start of a distinguished run of letters $+$, and divide by the r choices of that run. The normalization of the periodic measure cancels in the ratio. In particular $S^{\mathrm{P}}_{N,1} = Nu^2$, which gives the adjacent crossing $J = (\log N)/4$ of Stepanyan et al. [16]. Every interior ratio increases strictly from 0 to ∞ in $u > 0$, and the ratios increase strictly toward the middle, apart from the symmetric middle tie, by the coefficient comparison of Proposition 4.1.

Theorem 5.1 (periodic relative error)**.** *For $N \ge 2$, $1 \le k \le N-1$, and $u > 0$, define*

$$\widehat{S}^{\mathrm{P}}_{N,k}(u) = \frac{Nu}{\sqrt{k(N-k)}} I_1\Big(2u\sqrt{k(N-k)}\Big).$$

Then

$$0 \le 1 - \frac{S^{\mathrm{P}}_{N,k}(u)}{\widehat{S}^{\mathrm{P}}_{N,k}(u)} \le \frac{Nu^2}{2}. \tag{34}$$

So under periodic boundaries the ratio of every bin to an endpoint is close to a single Bessel function I_1, with the same kind of error as in Theorem 4.3.

Proof. Put $x = su$, $A = ab/N$, $U_r = U_r(a)$ and $V_r = U_r(b)$. Shifting the index in (33) gives $S^{\mathrm{P}}_{N,k}(u) = (Nu/s) \sum_{r \ge 0} U_r V_r x^{2r+1}/(r!(r+1)!)$, so the ratio to $\widehat{S}^{\mathrm{P}}_{N,k}$ is an average of the $U_r V_r$ with weights $x^{2r+1}/(r!(r+1)!)$, whose sum is $I_1(2x)$. By (15), $0 \le 1 - U_r V_r \le r(r+1)/(2A)$, and the second identity of Lemma 3.2(c) bounds the deficit by $x^2/(2A) = Nu^2/2$. □

We first give the idea behind the next theorem. A cyclic word switches an even number of times (it must come back to its first letter), so only I_1 appears in the periodic approximation, while the free words that start and end on different letters give the extra I_0 term of the free one. As $I_0 \approx I_1$ for large arguments, the free ratio is about twice the periodic one near the crossing. Both grow like e^{Nu}, so the periodic crossing needs Nu about $\log 2$ larger. Antal, Droz and Rácz [4, Eqs. (37) and (47)] work at a fixed value of $\tau = Nq$ (their $\theta = Nt/2$ with $t = 1 - \tanh J$). There the endpoint atoms are $1/(2\cosh\tau)$ (periodic) and $e^{-\tau}/2$ (free), which again differ by a factor of about 2 for large τ, while their densities at zero magnetization agree to leading order as $\tau \to \infty$.

Theorem 5.2 (periodic crossing and boundary shift)**.** *Let u_N^{P} be the positive root of $S_{N,m_N}^{\mathrm{P}}(u) = 1$, let $p_N^{\mathrm{P}} = 1/(1+u_N^{\mathrm{P}})$, and let $u_N^{\mathrm{F}} = u_N$ be the free central root. Put $c_{\mathrm{P}} = \frac{1}{2}\log(\pi/2)$. As $N \to \infty$,*

$$Nu_N^{\mathrm{P}} = L - \tfrac{1}{2}\log L + c_{\mathrm{P}} + \frac{\frac{1}{4}\log L - \frac{1}{2}c_{\mathrm{P}} + \frac{3}{8}}{L} + O\!\left(\frac{(\log L)^2}{L^2}\right), \tag{35}$$

$$N(u_N^{\mathrm{P}} - u_N^{\mathrm{F}}) = \log 2 + \frac{\frac{1}{4} - \frac{1}{2}\log 2}{\log N} + o(1/\log N). \tag{36}$$

Replacing Nu_N^{P} by $N(1-p_N^{\mathrm{P}})$ preserves (35). At the periodic crossing the coupling $J_N^{\mathrm{P}} = -\frac{1}{2}\log u_N^{\mathrm{P}}$ satisfies

$$J_N^{\mathrm{P}} = \tfrac{1}{2}(\log N - \log\log N) + \frac{\frac{1}{4}\log\log N - \frac{1}{2}c_{\mathrm{P}}}{\log N} + O\!\left(\frac{(\log\log N)^2}{(\log N)^2}\right). \tag{37}$$

Since $c_{\mathrm{P}} = c_0 + \log 2$, the periodic expansion is the free one shifted by $\log 2$, with $\frac{3}{8}$ in place of $\frac{1}{8}$ in the $1/L$ term.

Proof. Step 1. The first order. At the middle $2s = N + O(1/N)$. At $u = (1 \pm \theta)L/N$ with fixed $\theta \in (0,1)$ the bound of Theorem 5.1 tends to 0, and the large-argument form of I_1 makes $\widehat{S}^{\mathrm{P}}$ tend to ∞ with the plus sign and to 0 with the minus sign. Since S_{N,m_N}^{P} increases in u, we get $Nu_N^{\mathrm{P}} \sim L$.

Step 2. The crossing equation. Write $z = Nu_N^{\mathrm{P}}$. At the root $S^{\mathrm{P}} = 1$, so $\log \widehat{S}^{\mathrm{P}} = O(L^2/N)$ by (34). Replacing the argument $2su$ by z, and the factor Nu/s by $2z/N$, changes $\log \widehat{S}^{\mathrm{P}}$ by $O(L/N^2)$, and $I_1(z) = e^z(2\pi z)^{-1/2}(1 - 3/(8z) + O(z^{-2}))$ [18, Eq. 10.40.1]. So

$$z + \tfrac{1}{2}\log z = L + c_{\mathrm{P}} + \frac{3}{8z} + O(z^{-2} + L^2/N),$$

and Lemma 4.6 with $\ell = L$, $C = c_{\mathrm{P}}$ and $B = \frac{3}{8}$ gives (35). As $N(u - u/(1+u)) = O(L^2/N)$ for $Nu = O(L)$, it also holds for $N(1 - p_N^{\mathrm{P}})$.

Step 3. The shift and the coupling. By (2), $Nu_N = z_N + O(L^2/N)$. Subtracting (11), where $c_0 = c_{\mathrm{P}} - \log 2$ and the correction is $\frac{1}{8}$, gives (36), even with error $O((\log L)^2/L^2)$. Finally $J = -\frac{1}{2}\log u = \frac{1}{2}(L - \log z)$, and $\log z = \log L - (\frac{1}{2}\log L - c_{\mathrm{P}})/L + O((\log L)^2/L^2)$ gives (37). □

Before we return to the coupling (37) at the end of this section, we describe the whole row near the crossing. The factor e^{-2y^2} in the next theorem has a simple heuristic. Moving $yN/\sqrt{\log N}$ away from the middle multiplies the Bessel argument by $2\sqrt{k(N-k)}/N \approx 1 - 2y^2/\log N$. The argument is about $\log N$, so it drops by about $2y^2$, and the ratio, which grows like e to the argument, drops by about the factor e^{-2y^2}.

Theorem 5.3 (spatial crossing window)**.** *Let $X \in \{\mathrm{F}, \mathrm{P}\}$ and let u_N^X be its exact central crossing. For fixed real t, y set*

$$u = u_N^X + \frac{t}{N}, \qquad k_N(y) = \operatorname{nint}\left(\frac{N}{2} + \frac{yN}{\sqrt{\log N}}\right),$$

where either nearest-integer convention may be used. Then, as $N \to \infty$, uniformly on compact sets of (t, y),

$$S^X_{N,k_N(y)}(u_N^X + t/N) \longrightarrow e^{t-2y^2}. \tag{38}$$

If $M_N^X(t)$ is the number of interior bins with probability strictly above the endpoint probability at that parameter, then

$$\frac{\sqrt{\log N}}{N} M_N^X(t) \longrightarrow \sqrt{2\max(t, 0)}. \tag{39}$$

So the crossing window has width about $1/N$ in u and about $N/\sqrt{\log N}$ in space, and inside it the ratio of each bin to an endpoint is a Gaussian bump times e^t. A weak limit theorem (a limit for the overall shape of the law) cannot give this, since it says nothing about a single bin such as the endpoint.

Proof. Step 1. The profile. Put $\lambda = k/N - \frac{1}{2}$ and $\eta = 2\sqrt{k(N-k)}/N = \sqrt{1-4\lambda^2}$. For the stated rounding and bounded y,

$$\eta = 1 - \frac{2y^2}{L} + O\Big(L^{-2} + \frac{1}{N\sqrt{L}}\Big), \qquad N u_N^X = L + O(\log L).$$

So the Bessel argument $Nu\eta$ at (u, k) is $Nu_N^X + t - 2y^2 + o(1)$, uniformly on the compact set. By Theorems 4.3 and 5.1 both ratios are within relative error $O(L^2/N)$ of their Bessel forms, and $I_0(z)$ and $I_1(z)$ equal $e^z(2\pi z)^{-1/2}(1 + O(1/z))$ [18, Eq. 10.40.1], so the prefactors relative to their central values tend to 1 uniformly. The change of argument gives e^{t-2y^2}, and the exact central ratio is 1, which proves (38).

Step 2. The count. The interior ratios increase toward the middle, by Proposition 4.1 for $X = \mathrm{F}$ and by the coefficient comparison after (33) for $X = \mathrm{P}$. If $t \le 0$, the central ratio is at most 1, so no interior bin is above the endpoint. Let $t > 0$ and $0 < \theta < \sqrt{t/2}$. By Step 1, for large N the bins at scaled distance less than $\sqrt{t/2} - \theta$ are above the endpoint and those at distance $\sqrt{t/2} + \theta$ are below, and by monotonicity so are all farther bins. Count the bins in between, divide by $N/\sqrt{L}$ and let $\theta \downarrow 0$. Rounding changes the count by $O(1)$. □

Figure 4 shows the exact ratios at $N = 10^6$.

We end with the coupling at the two central crossings, since this is how Stepanyan et al. [16] state the crossing. Write $J_N^{\mathrm{F}} = -\frac{1}{2}\log u_N$ for the coupling at the free central crossing. For $J > 0$ let $\xi(J) = -1/\log\tanh J$ be the correlation length of the infinite chain, so that $\langle \sigma_0 \sigma_r \rangle = e^{-r/\xi(J)}$ there.

Corollary 5.4 (couplings and correlation lengths at the crossings)**.** *As $N \to \infty$,*

(a) The free coupling satisfies $J_N^{\mathrm{F}} = \frac{1}{2}(\log N - \log\log N) + \dfrac{\frac{1}{4}\log\log N - \frac{1}{2}c_0}{\log N} + O\Big(\dfrac{(\log\log N)^2}{(\log N)^2}\Big)$.

(b) The two couplings differ by $J_N^{\mathrm{F}} - J_N^{\mathrm{P}} = \dfrac{\log 2}{2\log N} + O\Big(\dfrac{\log\log N}{(\log N)^2}\Big)$.

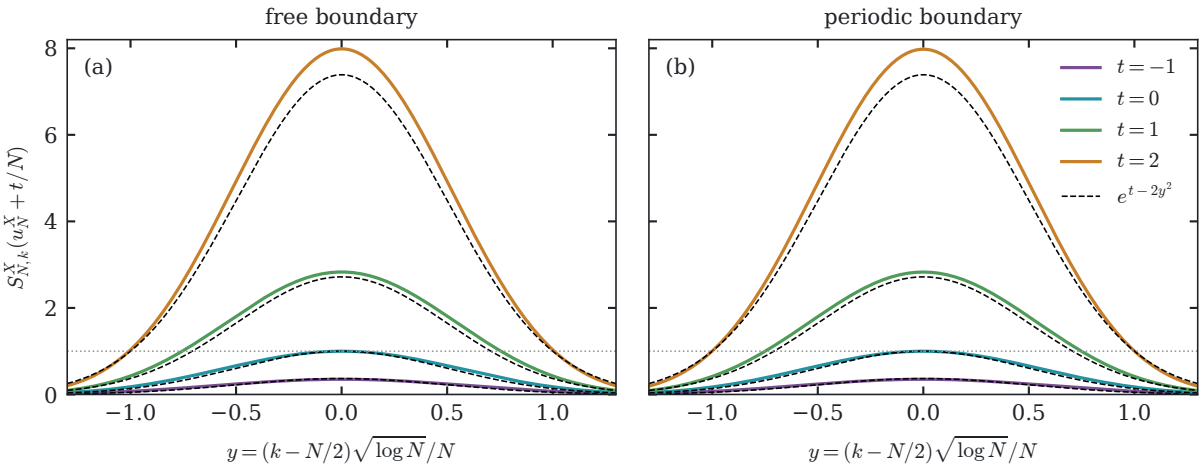


Figure 4: The exact ratios $S^X_{N,k}(u^X_N + t/N)$ at $N = 10^6$ for free (a) and periodic (b) boundaries, with the limit e^{t-2y^2} of Theorem 5.3 (dashed). Here $y = (k - N/2)\sqrt{\log N}/N$, and the bins above the dotted line are above the endpoint height. The convergence is slow, and at $t = 2$ and $y = 0$ the relative error is still 0.08.

(c) The correlation lengths satisfy $N/\xi(J^{\mathrm{F}}_N) = 2\log N - \log\log N + \log(\pi/8) + O(\log\log N/\log N)$, *and* $N/\xi(J^{\mathrm{P}}_N) = 2\log N - \log\log N + \log(\pi/2) + O(\log\log N/\log N)$.

All three parts are facts about Nu at the crossings, read on other scales. The coupling is $-\frac{1}{2}\log u$, so it sees Nu only through $\log(Nu)$. The shift of $\log 2$ in Nu between the two boundary conditions therefore becomes a shift of order $1/\log N$ in the coupling. The correlation length is $1/(2u)$ to leading order, so N/ξ is about $2Nu$.

Proof. (a) By (2), $u_N = z_N/(N - z_N)$, so $J^{\mathrm{F}}_N = \frac{1}{2}(L - \log z_N) + O(L/N)$. Taking logarithms in (11) gives $\log z_N = \log L - (\frac{1}{2}\log L - c_0)/L + O((\log L)^2/L^2)$, and (a) follows in the same way as (37) in Step 3 of the proof of Theorem 5.2.

(b) The difference is $J^{\mathrm{F}}_N - J^{\mathrm{P}}_N = \frac{1}{2}\log(u^{\mathrm{P}}_N/u^{\mathrm{F}}_N) = \frac{1}{2}\log(1+\delta)$, where $\delta = N(u^{\mathrm{P}}_N - u^{\mathrm{F}}_N)/(Nu^{\mathrm{F}}_N)$. By (36) the numerator is $\log 2 + O(1/L)$. By (2) and Corollary 3.7(b) the denominator is $z_N + O(L^2/N) = L + O(\log L)$. So $\delta = (\log 2)/L + O(\log L/L^2)$, and $\frac{1}{2}\log(1+\delta) = \delta/2 + O(\delta^2)$.

(c) With $u = e^{-2J}$ we have $\tanh J = (1-u)/(1+u)$, so $N/\xi(J) = N\log\frac{1+u}{1-u} = 2Nu + O(Nu^3)$ for $0 < u \le \frac{1}{2}$. At either crossing $Nu^3 = O(L^3/N^2)$. It remains to insert $Nu_N = z_N + O(L^2/N) = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$, from Corollary 3.7(b), and the expansion (35) of Nu^{P}_N, and to recall that $2c_0 = \log(\pi/8)$ and $2c_{\mathrm{P}} = \log(\pi/2)$. □

So at either crossing the correlation length is about $N/(2\log N)$. The chain is ordered over long stretches, but each stretch is a vanishing fraction of the chain, and N/ξ grows like $2\log N$. Antal, Droz and Rácz [4] give the magnetization law when τ is fixed, and there N/ξ stays bounded (it is about 2τ). The crossing lies outside that regime, which is why we need bounds that are uniform in τ. Part (b) is the shift by $\log 2$ in Nu of (36), seen on the logarithmic scale of the coupling. So the boundary condition moves the coupling at the crossing by an amount that decays only like $1/\log N$.

Stepanyan et al. [16, Section III.B] call the periodic central crossing the second thermal transition of the finite chain. They estimate it analytically by $\beta J \approx (2\log N - 1)/5$ [16, Eq. (16)], and their Fig. 1(b) shows the fitted line $\beta J \approx 0.3864\log N - 0.21055$. Table 3 and Figure 5

N	J_N^{P}	(37)	Eq. (16)	Fig. 1(b)	J_N^{F}	$J_N^{\mathrm{F}} - J_N^{\mathrm{P}}$
10	0.687	0.776	0.721	0.679	0.823	0.136
10^2	1.574	1.597	1.642	1.569	1.658	0.084
10^3	2.534	2.541	2.563	2.459	2.591	0.057
10^4	3.540	3.543	3.484	3.348	3.582	0.042
10^5	4.576	4.578	4.405	4.238	4.610	0.033
10^6	5.633	5.634	5.326	5.128	5.661	0.028

Table 3: The couplings J_N^{P} and J_N^{F} at the periodic and free central crossings, from the exact run polynomials, against the main terms of (37) and the two approximations of Stepanyan et al. [16] (their Eq. (16) and the line of their Fig. 1(b)). By Corollary 5.4(b) the last column is $(\log 2)/(2 \log N)$ to leading order, which is 0.030 at $N = 10^5$. Differences are computed before rounding.

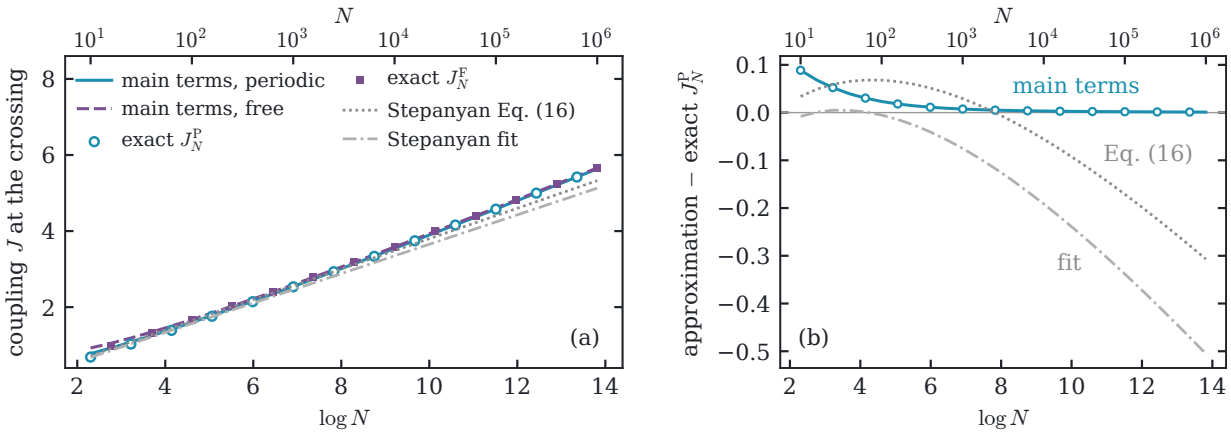


Figure 5: (a) The coupling at the central crossing against $\log N$, exact for periodic (circles) and free (squares) ends, with the main terms of (37) and of Corollary 5.4(a) and the two approximations of Stepanyan et al. [16]. (b) Each approximation of the periodic coupling minus the exact value. The main terms converge to the exact value. The two approximations have no $\log \log N$ term and a slope in $\log N$ below the limiting slope $\frac{1}{2}$, so they drift away.

compare both with the exact crossing and with (37) without its error term. The line is accurate for $10 \le N \le 100$, but neither approximation has a $\log \log N$ term. The main terms of (37) have slope $\frac{1}{2} - 1/(2 \log N)$ in $\log N$, up to $O(\log \log N/(\log N)^2)$, so the exact curve bends slowly toward slope $\frac{1}{2}$. Its secant slope is 0.385 on $10 \le N \le 10^2$, close to the fitted 0.3864, then 0.417 on $10^2 \le N \le 10^3$, and 0.450 on $10^4 \le N \le 10^5$. From $N = 10^3$ on, the main terms of (37) are closer to the exact value than either approximation, and at $N = 10^5$ they are off by 0.002, against 0.171 and 0.338.

6 Verification

Aside from Remark 3.9, which is computer-assisted throughout, the proofs use two computations, namely the integer certificates for $N \le 174$ in Theorem 3.4 (Step 4) and Proposition 3.8, and the evaluation at $N = 2774$ in Corollary 3.7(d). Each one is described where it is used. The scripts are standard-library Python. They check the exact law, the orderings, the Bessel bounds, the

asymptotic formulas and the couplings of Table 3, and where possible each reported number is computed in two independent ways.

Many of the results are formalized in Lean 4, in the folder `kagey-problems/problem131/lean/PaperA`. This includes Theorem 3.4 for all $N \geq 2$. For $N \leq 174$ the Lean kernel checks the same certificates as Step 4 of its proof, with rational bounds on ζ_N that Lean proves itself. Only the standard axioms are used. Not formalized are Proposition 3.8, Corollary 5.4, the remarks, and the terms after the leading one in the asymptotic results. The file `MAP.md` in that folder gives the Lean theorem for each result, or says that there is none.

Data and code availability. Everything is in the folder `kagey-problems/problem131` of https://github.com/apemm/Kagey-Problems, at the release `paperA-arxiv-v2`. To rerun the checks, run `python -B verify_all.py --paper a` in that folder. The file `SUPPLEMENT.md` says what each script does. Theorem 3.4's certificates are listed in `paperA/data/every_n_certificates.csv`, and `paperA/verify_ising_coupling.py` prints Table 3. Any number in Sections 3 and 5 that no script prints is listed in `paperA/data/quoted_values.txt`.

Acknowledgments. I thank Peter Kagey for posing the problem. I used two AI tools in this work, Claude Opus 5.5 (Anthropic, through Claude Code, in September and October 2026) and GPT-6 Astra (OpenAI). Except for the proofs of Theorem 3.4 and Corollary 5.4, I wrote the proofs and derivations of this paper by hand, without AI assistance, before any revision. I then used Claude Opus 5.5 to check everything I had written for typos and for statements that were technically wrong. It found a fair number of gaps, which I fixed, along with occasional wrong references and unclear wording. I decided every revision and went over each one by hand. The model also drafted rewordings of many passages, which I reviewed and edited. I had conjectured a bracket of the same kind for the crossing at every N, but it had gaps. Near the end of the work the model proposed an improved conjecture, which is Theorem 3.4 as stated here, together with a proof that uses the Bessel bounds of Lemma 3.6. I rechecked and rederived its proof by hand. In a later revision, which recast the introduction in terms of the Ising chain, the model proposed Corollary 5.4 and the comparison in Table 3, and wrote their proofs and the script that computes the table. I checked both by hand. I also checked the proofs, except those of Corollary 5.4, with GPT-6 Astra, which found nothing beyond the earlier check.

The Lean 4 formalization in `kagey-problems/problem131/lean/PaperA` was generated by the model, then revised and checked by hand against the corresponding results (see `MAP.md` in that folder). Parts of the Python verification scripts were also written by the model, which cross-checked them against each other. Lastly, I used it for literature search, namely to find and verify sources beyond the ones I had first checked on my own.

All mathematical claims, proofs and computations in this paper are my responsibility.

A The crossing at every N

This appendix proves Lemma 3.6, the analytic half of Theorem 3.4, the identity behind the certificates and Proposition 3.8. The letters U, U', W, W', r, r', t, t', τ, κ, m, B, D, β, Δ and δ are local to this appendix (each is defined where it first shows up). We round every decimal constant in the safe direction, namely so that the inequality it enters stays true.

A.1 Proof of Lemma 3.6

For integer n, $I_n(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}\cos(nt)\,dt$ [18, Eq. 10.32.3], so $I_0(z)+I_1(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}(1+\cos t)\,dt$. Substitute $s = 1-\cos t$, so that $\sin t = \sqrt{s(2-s)}$ and $1+\cos t = 2-s$, and then $s = v/z$. This gives

$$\mathcal{E}(z) = \frac{1}{\sqrt{\pi}}\int_0^{2z} e^{-v}v^{-1/2}\sqrt{1-\frac{v}{2z}}\,dv. \tag{40}$$

On $[0,1]$ we have $\sqrt{1-y} = 1-\sum_{k\ge1}c_k y^k$ with every $c_k>0$, $c_1=\frac12$ and $c_2=\frac18$, and $\sum_k c_k = 1$ by monotone convergence as $y\uparrow 1$. So $\sum_{k\ge3}c_k = \frac38$, and $y^k\le y^3$ gives $\sum_{k\ge3}c_k y^k \le \frac38 y^3$. Hence, for $0\le y\le 1$,

$$g_-(y) := 1-\frac{y}{2}-\frac{y^2}{8}-\frac{3y^3}{8} \le \sqrt{1-y} \le 1-\frac{y}{2}-\frac{y^2}{8} \le g_+(y) := 1-\frac{y}{2}-\frac{y^2}{8}+\frac{y^3}{8}.$$

The cubic g_- decreases and $g_-(1)=0$, so $g_-\le 0$ on $[1,\infty)$. Also $8g_+(y) = (y-2)^2(y+1)+2(y-1)^2+2>0$ for $y\ge -1$. Put $y = v/(2z)$ in (40), replace $\sqrt{1-y}$ by $g_\pm$ and extend the integral to ∞. The added piece is at most 0 with g_- and at least 0 with g_+, so the lower bound survives with g_- and the upper with g_+. Since $\frac{1}{\sqrt{\pi}}\int_0^\infty e^{-v}v^{k-1/2}\,dv = \Gamma(k+\frac12)/\Gamma(\frac12)$ equals $1, \frac12, \frac34, \frac{15}{8}$ for $k=0,1,2,3$, the two bounds integrate to the two sides of the lemma. The bound $\mathcal{E}\le 1$ follows in the same way from $\sqrt{1-y}\le 1$. Finally $-\frac{3}{128z^2}+\frac{15}{512z^3}\le 0$ exactly when $z\ge\frac54$.

A.2 Proof of Theorem 3.4 for $N \ge 175$

Step 1. Bounds that hold for every $\zeta\ge 4$. Fix $N\ge 175$ and write $\zeta=\zeta_N$, $U=\zeta+\frac{1}{8\zeta}$, $r=U/N$ and $n(\zeta) = \sqrt{8\zeta/\pi}\,e^\zeta$. Then $n(\zeta_N)=N$ by (8), and $\zeta\ge 4$ since $n(4)<175$. For $k=1,2,3$ the function $U^k/n(\zeta)$ decreases on $[4,\infty)$, because its logarithmic derivative is $k(1-\frac{1}{8\zeta^2})/U - 1 - \frac{1}{2\zeta} < \frac{k}{U}-1<0$. The same holds for U^4/n, since $U>4$, and for ζ/n and ζ^2/n. The values at $\zeta=4$ give, for every $\zeta\ge 4$,

$$r\le 0.023135,\quad \frac{U^2}{N}\le 0.093262,\quad \frac{U^3}{N}\le 0.375960,\quad \frac{U^4}{N}\le 1.51559,\quad \frac{\zeta(\zeta+1)}{N}\le 0.114776.$$

Step 2. The upper bound. Let $W = U/(1-r)$, so that $NW/(N+W)=U$, and let $\delta = W(W+2)/(2N)$. By Lemma 3.5(b) and (9) it suffices to show that $G_U := \psi(W)-\psi(\zeta)+\log\mathcal{E}(W)+\log(1-\delta)>0$. We bound the three parts from below. First, $\psi(W)-\psi(\zeta) = (W-U)+(U-\zeta)+\frac12\log\frac{U}{\zeta}+\frac12\log\frac{W}{U}$, where $W-U=\frac{Ur}{1-r}$, $U-\zeta=\frac{1}{8\zeta}$, $\log\frac{U}{\zeta}=\log(1+\frac{1}{8\zeta^2})\ge \frac{1}{8\zeta^2}-\frac{1}{128\zeta^4}$ and $\log\frac{W}{U} = -\log(1-r)\ge r$. Second, Lemma 3.6 gives $\mathcal{E}(W)\ge 1-t$ with $t=\frac{1}{8W}+\frac{3}{128W^2}+\frac{45}{512W^3}$. Since $W\ge U\ge\zeta$ and $\frac{1}{8U} = \frac{1}{8\zeta}-\frac{1}{8\zeta(8\zeta^2+1)}\le \frac{1}{8\zeta}-\frac{1}{64\zeta^3}+\frac{1}{512\zeta^5}$, we get $t\le\tau := \frac{1}{8\zeta}+\frac{3}{128\zeta^2}+\frac{37}{512\zeta^3}+\frac{1}{512\zeta^5}$ and $\log\mathcal{E}(W)\ge\log(1-\tau)\ge -\tau-\frac{\tau^2}{2(1-\tau)}$. Third, $\delta = \frac{Ur}{2(1-r)^2}+\frac{r}{1-r}$ and $\log(1-\delta)\ge -\delta-\frac{\delta^2}{2(1-\delta)}$. Adding the three parts gives $G_U\ge B+D$, where B collects the terms in ζ alone and D the terms with a factor r,

$$B = \frac{5}{128\zeta^2}-\frac{37}{512\zeta^3}-\frac{1}{256\zeta^4}-\frac{1}{512\zeta^5}-\frac{\tau^2}{2(1-\tau)},\qquad D = \frac{r}{2(1-r)}\Big(\frac{U(1-2r)}{1-r}-1-r\Big)-\frac{\delta^2}{2(1-\delta)}.$$

Both $\zeta\tau$ and τ decrease in ζ, so $\zeta^2 B$ increases. At $\zeta=4$ we have $\zeta\tau = 0.135384$, $\tau = 0.033846$ and $\zeta^2 B = 0.011236$. Hence $B>0$. In D the bracket is at least $4.03125(1-0.023683)-1.023135 > 2.912$, so the first term of D is at least $1.456\,r$. Write $\delta = rm$ with $m = \frac{U}{2(1-r)^2}+\frac{1}{1-r}$. Then $\delta\le 0.07255$

and $rm^2 = \frac{U^3}{4N(1-r)^4} + \frac{U^2}{N(1-r)^3} + \frac{r}{(1-r)^2} \le 0.2276$, so $\frac{\delta^2}{2(1-\delta)} = r \cdot \frac{rm^2}{2(1-\delta)} \le 0.1228\,r$. Hence $D \ge 1.33\,r > 0$. So $G_U > 0$ and $z_N < U$.

Step 3. The lower bound. Let $\varepsilon = \varepsilon_N$, $U' = U - \varepsilon$, $r' = U'/N$ and $W' = U'/(1-r')$, so that $NW'/(N+W') = U'$. By Step 1, $\varepsilon \le 0.114776 + \frac{1}{256} < 0.1187$, so $U' \ge \zeta - 0.1187 \ge 3.88$ and $W' \ge U'$. By Lemma 3.5(a) and (9) it suffices to show that $G_{U'} := \psi(W') - \psi(\zeta) + \log \mathcal{E}(W') < 0$. We bound the two parts from above. First, $\psi(W') - \psi(\zeta) = (U' - \zeta) + \frac{U'r'}{1-r'} + \frac{1}{2}\log\frac{U'}{\zeta} - \frac{1}{2}\log(1-r')$, where $U' - \zeta = \frac{1}{8\zeta} - \varepsilon$, $\frac{1}{2}\log\frac{U'}{\zeta} \le \frac{1}{2}(\frac{U'}{\zeta} - 1) = \frac{1}{16\zeta^2} - \frac{\varepsilon}{2\zeta}$ and $-\frac{1}{2}\log(1-r') \le \frac{r'}{2(1-r')}$. The two terms in r' add up to $\frac{U'^2 + U'/2}{N(1-r')}$. Second, Lemma 3.6 gives $\mathcal{E}(W') \le 1 - t'$ with $t' = \frac{1}{8W'} + \frac{3}{128W'^2} - \frac{15}{512W'^3} \in (0,1)$, so $\log \mathcal{E}(W') \le -t'$. Here $\frac{1}{8W'} = \frac{1}{8U'} - \frac{1}{8N}$. Write $U' = \zeta(1+\kappa)$ with $\kappa = \frac{1}{8\zeta^2} - \frac{\varepsilon}{\zeta}$. Then $\frac{1}{U'} \ge \frac{1-\kappa}{\zeta}$, so $-\frac{1}{8W'} \le -\frac{1}{8\zeta} + \frac{1}{64\zeta^3} - \frac{\varepsilon}{8\zeta^2} + \frac{1}{8N}$. Next, $x \mapsto x/(1-x/N)$ increases and $U' \le U$, so $W' \le U/(1-r)$ and $\frac{1}{W'^2} \ge \frac{(1-r)^2}{\zeta^2(1+\frac{1}{8\zeta^2})^2} \ge \frac{(1-2r)(1-\frac{1}{4\zeta^2})}{\zeta^2}$. Finally $\frac{15}{512W'^3} \le \frac{15}{512U'^3}$. Adding, and using $\varepsilon(1+\frac{1}{2\zeta}) = \frac{\zeta^2 + \frac{3}{2}\zeta + \frac{1}{2}}{N} + \frac{1}{16\zeta^2} + \frac{1}{32\zeta^3}$, we get $G_{U'} \le \beta + \Delta - \frac{\varepsilon}{8\zeta^2}$ with

$$\beta = -\frac{3}{128\zeta^2}(1-2r)\Big(1 - \frac{1}{4\zeta^2}\Big) + \frac{15}{512U'^3} - \frac{1}{64\zeta^3}, \qquad \Delta = \frac{1}{N}\Big(\frac{U'^2 + U'/2}{1-r'} + \frac{1}{8} - \zeta^2 - \frac{3}{2}\zeta - \frac{1}{2}\Big).$$

For $\beta < 0$ it suffices that $\zeta^2/U'^3 \le \frac{4}{5}(1-2r)(1-\frac{1}{4\zeta^2})$. The left side is at most $\zeta^2/(\zeta - 0.12)^3 \le 16/3.88^3 < 0.274$, since this function decreases for $\zeta > 0.12$ (its logarithmic derivative is $\frac{2}{\zeta} - \frac{3}{\zeta - 0.12} < 0$ there). The right side is at least $0.8 \cdot 0.9537 \cdot 0.9843 > 0.75$. For $\Delta < 0$, note that $U' \le U$ and $r' \le r$, so

$$\frac{U'^2 + U'/2}{1-r'} \le U^2 + \frac{U}{2} + \frac{U^3/N + U^2/(2N)}{1-r} \le \zeta^2 + \frac{\zeta}{2} + 0.2667 + 0.4326,$$

using $U^2 + \frac{U}{2} = \zeta^2 + \frac{\zeta}{2} + \frac{1}{4} + \frac{1}{16\zeta} + \frac{1}{64\zeta^2}$. This is less than $\zeta^2 + \frac{3}{2}\zeta + \frac{3}{8}$ because $\zeta \ge 4$. Hence $G_{U'} < 0$ and $z_N > U'$.

The constants were recomputed in 40-digit arithmetic, and as a check outside the proof we evaluated $\zeta^2 B$, D, β and Δ on $\zeta \in [4, 60]$ with step 0.001.

A.3 The certificates

Let $c, d > 0$ be integers. Multiplying (6) by d^{N-1} gives

$$d^{N-1} S_N(c/d) = \sum_j W(N, m_N, j)\, c^j d^{N-1-j},$$

an integer. By Theorem 2.1 it is the sum of $d^{\mathrm{rep}(w)} c^{\mathrm{chg}(w)}$ over the words w in bin m_N, that is, the numerator $n_N(m_N)$ of Remark 2.3 with $(\alpha, \beta) = (d, c)$. The recursion (3) with the weights d, c in place of p, q and initial values 1 computes the same sums. So $S_N(c/d) < 1$ exactly when this integer is less than d^{N-1}. This is the identity used in Step 4 of the proof of Theorem 3.4.

A.4 Proof of Proposition 3.8

Step 1. An exact formula for R_N. Write $w = w_N$, $z = z_N$, $\Lambda_N = 2\Phi(w)/N$ and $\delta = w(w+2)/(2N)$. At $w = w_N$ the display in the proof of Lemma 3.5 gives $1 \le \Lambda_N \le 1/(1-\delta)$ when $\delta < 1$, and (9)

gives $\psi(w)+\log\mathcal{E}(w)=L+c_0+\log\Lambda_N$. Since $w-z=wz/N$, $w/z=1+w/N$ and $\frac{1}{8z}=\frac{1}{8w}+\frac{1}{8N}$, the remainder $R_N=\psi(z)-L-c_0-\frac{1}{8z}$ is

$$R_N=\log\Lambda_N-\Big(\log\mathcal{E}(w)+\frac{1}{8w}\Big)-\frac{1}{8N}-\frac{wz}{N}-\frac{1}{2}\log\Big(1+\frac{w}{N}\Big). \tag{41}$$

Step 2. The lower bound for $N\ge 175$. By the proof of Theorem 3.4, $3.88\le z\le w\le U/(1-r)$. Note that $\log\Lambda_N\ge 0$ and $-\log\mathcal{E}(w)\ge -\log(1-\frac{1}{8w})\ge\frac{1}{8w}$ by Lemma 3.6, since $w\ge\frac{5}{4}$. So $R_N\ge -\frac{wz+w/2+1/8}{N}=-\frac{z^2+z/2}{N-z}-\frac{1}{8N}$. This is at least $-\frac{z^2+z}{N}$ as soon as $N\ge\frac{2z^2(z+1)}{z-1/4}$. The right side increases in z and $z\le U$, so it suffices that $U^2/N\le\frac{U-1/4}{2(U+1)}$. The left side is at most 0.0933 and the right side at least 0.3757.

Step 3. The upper bound for $N\ge 175$. Let $t=\frac{1}{8w}+\frac{3}{128w^2}+\frac{45}{512w^3}\le\frac{1.09504}{8w}$, which holds as $w\ge 3.88$. Lemma 3.6 and $-\log(1-t)\le t+\frac{t^2}{2(1-t)}$ give

$$-\log\mathcal{E}(w)-\frac{1}{8w}\le\frac{3}{128w^2}+\frac{45}{512w^3}+\frac{t^2}{2(1-t)}\le\frac{0.0559}{w^2}\le\frac{0.0559}{z^2}.$$

Using $\log\Lambda_N\le\delta+\frac{\delta^2}{2(1-\delta)}$, $\log(1+x)\ge x-\frac{x^2}{2}$ and $wz\ge w^2-w^3/N$, the rest of (41) is at most $-\frac{(w-1/2)^2}{2N}+\frac{w^3+w^2/4}{N^2}+\frac{\delta^2}{2(1-\delta)}$. By the bounds on r and U^k/N in Step 1 of the previous proof, with $w\le U/(1-r)$, we have $\frac{w^3+w^2/4}{N}\le 0.4278$ and $N\frac{\delta^2}{2(1-\delta)}=\frac{(w^4+4w^3+4w^2)/(4N)}{2(1-\delta)}\le 0.4945$, while $\frac{(w-1/2)^2}{2}\ge 5.71$. So the rest is negative and $R_N\le 0.0559/z^2<\frac{1}{16z^2}$.

Step 4. The form in L for $N\ge 175$. We have $z<U=L+c_0-\frac{1}{2}\log\zeta+\frac{1}{8\zeta}<L$. Also $z>\zeta-0.1187\ge\nu_N-0.1187\ge L/2$, because $L-\log L\ge 1.18$ for $L\ge\log 175$. Then $\frac{1}{16z^2}\le\frac{1}{4L^2}$ and $\frac{z^2+z}{N}\le\frac{L^2+L}{N}$.

Step 5. Short chains. Here $N\le 174$. The function $\phi(z)=\psi(z)-L-c_0-\frac{1}{8z}$ increases in z, and $R_N=\phi(z_N)$, while $\frac{1}{16z^2}$ and $-\frac{z^2+z}{N}$ decrease. So it suffices to check $\phi(z(c'/d))\le\frac{1}{16z(c'/d)^2}$ and $\phi(z(c/d))\ge-\frac{z(c/d)^2+z(c/d)}{N}$ with the certificates, together with $z(c/d)\ge L/2$ and, for $N\ge 3$, $z(c'/d)\le L$. All hold, and the smallest gaps are 0.0373 (upper) and 0.0813 (lower), both at $N=174$. For $N=2$ we have $S_2(u)=2u$, so $u_2=\frac{1}{2}$ and $z_2=\frac{2}{3}$, and $\frac{2}{3}<\log 2$ is equivalent to $e^2<8$, which holds since $e^2<2.72^2=7.3984$.

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